Errors in the Signals Table

`gld_player_signals_daily`: what is wrong, how it is defined, what could be done, and where to see it

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Scope. Brands 14, 23, 64, 72 and 73. September 2026, unless another period is given. All rows, no sampling, aggregates only. The calculations and the evidence of each point are in the notebooks `01_distributions`, `02_validation` and `03_three_errors` of `signal_audit/notebooks/`; the status of each column is at the end of part A of `02_validation`.

1 Summary

In one sentence. In the signals table, the amounts and counts pass their checks (99 of 165 columns), but it has serious errors in time of play, recency and active days, and one of them uses data from the future. Several platform scores, including `churn_score`, are built on these columns.

The five most serious errors

	Error	What it affects	What could be done
1	`days_since_bet` looks 7 days into the future	`churn_score`, `deposit_recency_score`, `lifecycle_stage` and any model that uses it (false AUC of 0.99)	Calculate it with data up to the date
2	`active_days_*` counts days played only with playable bonus	17% to 34% of recent players look active when they are not; `frequency_decline`, `engagement_score`	Count only real money and released bonus
4	`value_decline` has no upper limit (up to 57 million)	`churn_score`	Cap it at $[0, 100]$
5	Time of play is not updated on some days (9 days, all brands)	All `tp_*` columns	A freshness check in the daily process
9–12	Night, weekend and entropy on another scale (up to 4)	8% to 28% of the rows; `risk_score` through `rg_night_play_ratio`	Calculate them as shares from 0 to 1

Priority of each point. HIGH affects a platform score or a model today, in many rows. MEDIUM wrong in many rows but not used for decisions, or already fixed from now on. LOW few rows, only the history, or a format detail. NO DEFINITION cannot be checked because the workbook does not give the full definition.

HIGH	1 recency, 2 active days, 3 `churn_score`, 4 `value_decline`, 5 freshness of time of play
MEDIUM	6 session, 7 gap between sessions, 8 `play_secs`, 9 night, 10 `rg_night_play_ratio`, 11 weekend, 12 entropy, 13 `deposit_recency_score`, 14 tenure, 15 `recency_pressure`, 16 `brand_id`, 17 empty codes
LOW	19 `hold_pct_l30d`, 18 `top_peak_*`, 20 `deposit_frequency_score`, 21 units in the workbook, 22 tax at 0
NO DEFINITION	23 NGR, 24 `prior_*_score`, 25 z-scores

2 The 165 Columns

The table has **165 columns**. Each one has its checks (`signal_checks.yaml`): its formula in the workbook, an accounting rule, a window that must be the sum of its daily values, a `prior_*` value that must be the one of the window before, or a limit (an impossible value, or a longer window smaller than a shorter one). A check *passes* if it holds in at least 99% of the rows of two snapshots (15 September and 1 October 2026). Several columns were also compared with the daily tables or with the transactions; this is how the errors of definition show up.

Status		Columns
Wrong	Wrong	37
Affected	Affected by a wrong input	12
Unconfirmed	Unconfirmed (no full definition)	11
Pass	Pass (they pass their checks)	99
Not checkable	Not checkable (keys, text, metadata)	6
Total		165

- **Pass (99):** the ids, the windows of amounts and counts (stake, wins, GGR, deposits, withdrawals, cashout, bets, sessions), their `prior_*`, the accounting rules, and the scores that follow their formula on correct inputs. They can be used as they are.
- **Wrong (37)** and **affected (12):** points 1 to 22 of the list. A column can pass its checks and still be wrong: `active_days_*` is the sum of its daily value, but the daily value uses a wrong definition.
- **Unconfirmed (11):** the workbook does not give the full definition (points 23 to 25).

The list of the 165 columns, with their status, their point and their checks, is in annex B.

Notation. D: date of the row. S: stake in the window; S_X: stake of the bets that meet X. W_d^r, W_d^l, W_d^j: stake on day d with real money, released bonus and playable bonus. G: GGR. P: amount won by the player. Session s: start a_s, end b_s, n_s bets. **M053, M113, M127...**: code of the measure in the data model workbook (`gld-schemas.xlsx`, sheet *Measures*).

3 Evaluation Criteria

Every column is evaluated with the same criteria. This is the full list.

1. The data. Two snapshots of the table, 15 September and 1 October 2026; the five brands (14, 23, 64, 72 and 73); the latest version (`_version`) of each row; all the rows, no sampling. The windows and lags also use the days before each snapshot. The format, distribution and share checks use every day of September 2026.

2. The checks of the table itself. Each column has one or more in `signal_checks.yaml`. The table gives each kind, its rule and how many columns and checks use it; annex B gives the exact rule of each one.

Kind	What it checks	Rule to pass	Columns	Checks
Formula	The value is the formula of its workbook measure (M code) on other columns of the same row.	$\lvert x - f \rvert \le 0.01 + 0.001\,\lvert f \rvert$	18	18
Identity	An accounting relation between columns, for example stake = real money + bonus.	$\lvert x - f \rvert \le 0.01 + 0.001\,\lvert f \rvert$	30	30
Window	The window of w days is the sum of its daily value over those w days.	$x_w(D) = \sum_{d \in (D-w,\, D]} x_{\text{today}}(d)$	32	32
Lag	A `prior_*` column is the value of its column k days before.	$x^{\text{prior}}(D) = x(D-k)$	14	14
Limit	A condition that must always hold: possible values (for example 0 to 100) and the order between windows.	the condition is true, e.g. $x_{7d} \le x_{30d}$	80	85
Band	The band matches its score with the cutoffs of the workbook.	band j if $c_j \le s < c_{j+1}$	3	3
Validated	Compared player by player with its value recalculated from the daily tables.	same value	3	3
Format	The text of an id or a code follows its format: UUID, digits only, a code in upper case, never empty.	the condition is true	5	5
Distribution	A categorical column is not spread evenly; if its levels have an order (value, risk), they fall from the first to the last.	checked on its figure	9	9
Share	An amount is a part of another one: the tax, a part of the GGR.	$0 <$ tax $\le$ GGR	3	3
No check	Keys, text and metadata.	—	8	8
Total			165	210

x: the value of the column in the table. f: the expected value (the formula, the identity, the sum of the window or the value k days before). The tolerance $0.01 + 0.001\,\lvert f \rvert$ lets the rounding of amounts pass: one cent plus 0.1%. A column can have several checks (so there are 210 for 165 columns).

3. The share that passes. For each check, with the two snapshots together:

$$\text{pass} = 100 \cdot \frac{\#\{\text{rows that meet the rule}\}}{\#\{\text{rows with a value}\}}.$$

Rows with NULL do not count. A check *passes* if it holds in at least 99% of the rows; it is *almost* from 95% to 99%; and it *fails* below 95%.

4. Comparisons with recalculated data. A column can follow its own rules and still be wrongly defined: for example, `active_days_l30d` is the exact sum of its daily value, but the daily value uses another definition. So several columns are also compared with their recalculated value:

- **From each bet of the transactions:** mean and median session length, gap between sessions, `play_secs`, night, weekend and hourly entropy.
- **From the daily gaming table:** `days_since_bet`, `active_days_*` and `tenure_days`.

The correct definition used in each case is in its point (field *Definition*). What I recalculate from the transactions is compared as a distribution, not player by player: the `accountId` of the landing transactions is another identifier than the `player_id` of gold (2 of about 18,000 players match), for a reason I do not know; `partyId` is still to test.

5. Impossible and suspicious values. Each metric has a real-life range (`metrics.yaml`). *Impossible*: a value that cannot come from real data, for example a share above 1 or a session of more than 1 day. *Suspicious*: a value that is possible but very rare. They are counted per brand and day (section 8).

6. The status of each column.

- **Wrong**: it fails one of its checks, or the comparison of criterion 4 shows that it does not follow the workbook definition, even if it meets its rules.
- **Affected**: it follows its formula, but one of its inputs is wrong.
- **Unconfirmed**: the workbook does not give the full definition, so I cannot say if it is right.
- **Pass**: none of the cases above, and all its checks pass (the worst holds in 99.92% of the rows).
- **Not checkable**: technical keys and metadata (`_version`, `param_version`, `has_sufficient_data`), and the load times and the snapshot date (`_ingested_at`, `_processed_at`, `snapshot_date`), which cannot be checked with the lake files.

7. The priority of each point. HIGH it affects a platform score or a model today, in many rows. MEDIUM wrong in many rows but not used for decisions, or already fixed from now on. LOW few rows, only the history, or a format detail. NO DEFINITION the workbook does not give the definition.

4 High Priority

They affect a platform score or a model today, in many rows.

1. HIGH `days_since_bet`

Error

- **Uses future data:** it matches the recency of 7 days later ($D+7$) for 92% of the players.
- **Effect:** AUC of 0.99 to predict a bet in the next 7 days, against 0.79–0.87 with the real recency.
- **History:** 0 for almost every player until May 2026.

Definition

- **Table:** $R_D = (D+7) - \max\{d \le D+7 : \text{bet}\}$.
- **Correct:** $R_D = D - \max\{d \le D : \text{bet}\}$.

Proposal

- Calculate it with the daily activity up to D (`days_since_last_bet` of the pipeline).

Evidence Fig. 14, Fig. 15, Fig. 16.

2. HIGH `active_days_today`, `active_days_l7d`, `active_days_l30d`, `active_days_l90d`, `active_days_l180d`, `prior_active_days_*`

Error

- **It counts days played only with playable bonus**, which are not active days.
- **Evidence:** it matches a count of any wallet for 96%–99.7% of the players, and the official definition for only 48%–86%. It never counts fewer days.
- **Effect:** 17%–34% of the players who bet in 30 days have 0 active days, and the table gives them at least one.
- **Scope:** `active_days_today` too (the windows are its sum); `active_days_l180d` was not compared directly. Affects `frequency_decline` and `engagement_score`.

Definition

- **Table:** $A_w = \#\{d \in (D-w, D] : W_d^r + W_d^l + W_d^j > 0\}$.
- **Correct (M053):** $A_w = \#\{d \in (D-w, D] : W_d^r + W_d^l > 0\}$.

Proposal

- Count only the days with a stake from real money or released bonus.

Evidence Fig. 17, Fig. 18.

3. HIGH **`churn_score` (and `churn_propensity_raw`, `churn_band`)**

Error
- **It follows its formula, but with wrong inputs:** `days_since_bet` (point 1), the active days (point 2) and `value_decline` with no upper limit (point 4).
- **In practice:** its AUC (0.78–0.83) does not show the leak.

Definition
- **Formula:** $\text{clamp}_{[0,100]}(0.5\,R + 0.3\,F + 0.2\,M)$, with $R =$ `recency_pressure`, $F =$ `frequency_decline`, $M =$ `value_decline`.

Proposal
- Recalculate it with the recency up to the date, the official active days and M capped at $[0, 100]$, or do not use it.

Evidence Fig. 16, Fig. 20.

4. HIGH **`value_decline` (`monetary_score` in the workbook)**

Error
- **No upper limit:** it reaches 57 million in September (802 million in the history).
- **How much:** 0.84% of the rows are above 100.
- **Affects:** `churn_score`.

Definition
- **Workbook (M125) and table:** $M = 100\max\big(0, 1 - \frac{G_7/7}{G_{90}/90}\big)$; no limit when G_{90} is small or negative.
- **Correct:** $\min(100, M)$.

Proposal
- Cap it at $[0, 100]$ and treat $G_{90} \leq 0$ apart.

Evidence Fig. 20.

5. HIGH **All `tp_*`: coverage and freshness**

Error
- **Start:** 6 July 2026, with about 5 000 rows a day.
- **Changes:** on 1 August it goes to about 300 000 rows; on 19 August the session changes.
- **Frozen days:** 9 days are exactly the same as the day before (21–24 and 26 August; 4, 11, 14 and 15 October), in every brand.
- **Affects:** all `tp_*`.

Proposal
- Do not use the `tp_*` of the table in the history; use the recomputed ones.
- A freshness check in the daily process.

Evidence Fig. 13, Fig. 12.

5 Medium Priority

Wrong in many rows but not used for decisions, or already fixed from now on. The `tp_*` columns exist since 6 July 2026.

6. MEDIUM `tp_avg_session_duration_secs`, `tp_median_session_duration_secs`

Error

- **Sessions that never close:** 0.6% last more than 1 day, up to 28 days.
- **Sessions of one event:** 1.8% last exactly 1 s.
- **Outside 10 s to 4 h:** 8.4% (mean) and 6.6% (median); 14.1% in brand 73.
- **Other facts:** the window is 90 days, and the value is not `play_secs/sessions` (it matches in 2.9%).
- **Part are sports bets:** the players who bet on sports (providers 58 and 50) have 7 times more sessions over 1 day (0.57% against 0.08%) and hold 64% of the cases; the other 36% is casino.

Definition

- **Table (deduced):** $\bar{d} = \text{mean}_s(b_s - a_s)$ over 90 days, open sessions not closed.
- **Correct:** session = run of bets with no gap over 30 min; $\bar{d} = \text{mean}\{t_{\text{last}} - t_{\text{first}} : n_s \geq 2\}$; one-bet sessions counted apart.

Proposal

- Rebuild the sessions from each bet: only 0.3% stay outside the range (mean per player) and the median hardly changes (25 vs 28 min).
- Note: even then there are sessions of up to 30 h of non-stop play, probably automatic.
- Separate the sports bets (providers 58 and 50) before building the sessions: a sports bet stays open until the match ends.

Evidence Fig. 1, Fig. 2, Fig. 3, Fig. 4, Fig. 6, Fig. 5.

7. MEDIUM `tp_avg_session_gap_hours`

Error

- **Negative gaps:** 0.91% of the rows, down to −262 h; a session starts before the previous one ends.
- **Where:** every day and every brand, more in 72 and 73.
- **Not sports:** 60% of the negative gaps are of players who only play casino (0.70% of their rows).

Definition

- **Definition:** $g_s = a_{s+1} - b_s$.
- **Correct:** with the 30-minute cut, always $g_s > 30$ min.

Proposal

- Calculate the gap with the rebuilt sessions.

Evidence Fig. 4, Fig. 5.

8. MEDIUM `play_secs_today`, `play_secs_l7d`, `play_secs_l30d`, `play_secs_l90d`, `play_secs_l180d`

Error
- **More time than the day has:** 3.7% of the days with play go over 24 h (6.0% in brand 73).
- **Maximum:** 26 million seconds (305 days) in September; 210 million in the history.
- **Affects:** `engagement_score` and `session_intensity_score`.
- **Mostly sports:** the days with sports bets go over 24 h in 13.9% to 14.6% of the rows, against 1.05% of the casino-only days, and hold 77% of the cases. In casino, the error of adding up games played at the same time is left.

Definition
- **Table (hypothesis):** sum per game, $\sum_j \text{secs}_{j,d}$, so games played at the same time count twice.
- **Correct:** the union of the session intervals of the day, $\left|\bigcup_s [a_s, b_s] \cap d\right| \leq 86\,400$ s.

Proposal
- Calculate it with the rebuilt sessions of each day.
- Do not compute the time of play of sports bets: in sports it measures how long the bet is open.

Evidence Fig. 4, Fig. 5.

9. MEDIUM `tp_night_index`

Error
- **Not a share:** it reaches 4, with values on $4/k$ (4, 2, 4/3, 1…).
- **How much:** 8.1% of the rows are above 1.
- **Also:** the night hours or the time zone seem different (median 0.20 vs 0.41).

Definition
- **Table:** $\dfrac{S_{\text{night}}/S}{6/24}$, an index against an even spread.
- **Correct (M127):** $S_{\text{night}}/S \in [0, 1]$, from 00:00 to 06:00 local time.

Proposal
- Calculate it as a share, from each bet.

Evidence Fig. 8, Fig. 7, Fig. 9, Fig. 12.

10. MEDIUM `rg_night_play_ratio`

Error
- **It should be equal to** `tp_night_index`, but it only matches in 91.5% of the rows.
- **It looks like** a version capped at 1.
- **Affects:** `risk_score`.

Definition
- **Workbook:** S_{night}/S.
- **Table:** always in $[0, 1]$, but different from `tp_night_index`.

Proposal
- Replace it with the recomputed night share (point 9).

Evidence Annex B: it matches `tp_night_index` in 91.5% of the rows.

11. MEDIUM `tp_weekend_index`

Error
- **Not a share:** it reaches 3.5, with values on $3.5/k$.
- **How much:** 23.4% of the rows are above 1.

Definition
- **Table:** $\dfrac{S_{\text{weekend}}/S}{2/7}$.
- **Correct (M147):** $S_{\text{Saturday and Sunday}}/S \in [0, 1]$, local time.

Proposal
- Calculate it as a share. Multiplying the table by 2/7 already gives almost the same (median 0.40 vs 0.41).

Evidence Fig. 8, Fig. 7, Fig. 10.

12. MEDIUM `tp_hourly_entropy`

Error
- **Wrong scale:** it is in bits and not normalised; it reaches 4.56.
- **How much:** 27.8% of the rows are above 1 (the workbook says 0–1).
- **Evidence of the base:** exact masses at $\log_2 2 = 1$, $\log_2 3 = 1.585$, $\log_2 4 = 2$, and none at $\ln 2$ or $\ln 3$.

Definition
- **Table:** $H = -\sum_h p_h \log_2 p_h \in [0, 4.58]$.
- **Correct:** $H/\log_2 24 \in [0, 1]$, with $p_h = S_{\text{hour } h}/S$ in local time.

Proposal
- Divide by $\log_2 24$, or calculate it from each bet (median 0.53 vs 0.47).

Evidence Fig. 8, Fig. 7, Fig. 11.

13. MEDIUM `deposit_recency_score`

Error
- **It measures the wrong thing:** it is a deposit score, but it uses the days since the *last bet* (`days_since_bet`), not since the *last deposit*.
- **Example:** a player who bets every day but has not deposited for two months looks as if they deposited yesterday.
- **The error is in the workbook definition,** not in the table: the table follows that formula.
- **It also has the leak** of `days_since_bet` (point 1).

Definition
- **Workbook (M113) and table:** 100 if there is a deposit today; otherwise $100\,(1 - \min(1, \text{days_since_bet}/30))$.
- **Correct:** the same with the days since the last deposit.

Proposal
- Calculate the days since the last deposit with the payments table.

Evidence Annex B: it follows its formula; the error is in the input.

14. MEDIUM `tenure_days`

Error
- **In short:** until July 2026 it was wrong for about 8 in 10 players; since September it moves forward correctly.
- **Evidence:** between two snapshots it must grow by the days in between; until July only 11%–33% do. On 1 August it jumps 10–16 days too many.
- **Notes:** (1) the old snapshots are still wrong; (2) moving forward correctly does not mean the value is correct; (3) the churn model does not use it: it calculates its own.

Definition
- **Correct:** $T_D = D - \min\{d : \text{active day}\}$, so $T_{D_2} - T_{D_1} = D_2 - D_1$.

Proposal
- Calculate it with the daily activity up to the date.
- Compare the values since September with that tenure.

Evidence Fig. 19.

15. MEDIUM `recency_pressure` (`recency_score` in the workbook)

Error
- **Stuck at the top:** it is 100 in 91.8% of the rows, so it hardly separates players.
- **Type:** a design problem, not a calculation error.

Definition
- **Workbook:** $\min\big(100, 100\,\text{days_since_bet}/\max(\tilde{g}, 3)\big)$, with $\tilde{g}$ the typical gap; it reaches 100 after 3 days without a bet.

Proposal
- Use the ratio with no cap (days without a bet/$\tilde{g}$), or the recency in days.

Evidence Fig. 20.

16. MEDIUM `brand_id`

Error
- **Empty:** 100% until 5 July 2026, 99% in July and 31%–33% from 1 August.

Proposal
- Take the brand of the player from `gld_player_gaming_daily` or from the activity cache (each player has only one).

Evidence Fig. 40.

17. MEDIUM `country`, `language`, `currency`: empty

Error
- **Always empty:** `country` and `language` are empty text in 100% of the rows, in every brand.
- **Almost always empty:** `currency`, in 85% to 89% of the rows.
- **The format of what is there is right:** upper case, 3 letters (USDT, 4), no mix such as TR and tr.

Definition
- **Expected:** an ISO code in upper case, and NULL (not empty text) when it is not known.

Proposal
- Fill them from the players table, or leave NULL when it is not known.

Evidence Fig. 41.

6 Low Priority

Few rows, only the history, or a format detail.

18. LOW `top_peak_hour, top_peak_day`

Error
- **Unclear default value:** 0 in 75% and 77% of the rows.
- **Problem:** 0 is also midnight and a real day, so "no data" cannot be told apart.

Definition
- **Table:** $\arg\max_h S_{\text{hour } h}$, and 0 if $S = 0$.
- **Correct:** NULL if $S = 0$.

Proposal
- Leave it empty (NULL) when there is no data.

Evidence Fig. 20.

19. LOW `hold_pct_l30d`

Error
- **Over 100% (impossible):** 479 days of the history, up to 6551%; in the five brands, 232 rows from 17 June to 17 August 2026, almost all in brand 72.
- **Likely cause:** negative wins (`won_eur_l30d` down to −747 €).
- **Noise with few bets:** the mass at 100% is players who won nothing; 80% of them have 5 bets or fewer.
- **The bar at** −100% in the distributions is not a peak: it adds up every value below −100% (0.63% of the rows), which in fact spread down to −300% and beyond.

Definition
- **Definition:** $h = 100\,(S - P)/S$, so $h > 100$ only if $P < 0$.

Proposal
- Calculate it only with $S > 0$ and $P \geq 0$.

Evidence Fig. 21, Fig. 22, Fig. 23, Fig. 24.

20. LOW `deposit_frequency_score`

Error
- **Empty before 2026:** the payments table has almost no deposits before that.

Proposal
- Do not use it in the history.

Evidence Annex B.

21. LOW **Units of the scores in the workbook**

Error
- **The workbook says 0–1** for every score; the table uses 0 to 100 (so does the rule of M107).

Proposal
- Fix the unit in the workbook.

Evidence Fig. 20.

22. LOW **`tax_eur_*`: always 0**

Error
- **No tax:** the 27.1 million rows of September are 0 (not NULL) for the day, 90 and 180 days; the 7- and 30-day windows are their sum, also 0.
- **Result:** I cannot check that the tax is a logical share of the GGR.

Definition
- **Expected:** $0 < \text{tax} \leq \text{GGR}$, a fixed share of the GGR (for example 10% to 20%).

Proposal
- Confirm whether these brands have a tax on the stake; if not, document it; if they do, the source does not send it.

Evidence Annex B.

7 Pending the Definition (Unconfirmed)

The workbook does not give the full definition, so they cannot be checked.

23. NO DEFINITION **`ngr_eur_today`, `ngr_eur_l7d`, `ngr_eur_l30d`, `ngr_eur_l90d`, `ngr_eur_l180d`**

Error
- **It matches** in 79% (180 days) to 99% (today) of the rows; the table is lower.
- **Its windows are** the sum of its daily value.

Definition
- **Workbook:** "GGR minus the bonus cost and the taxes", with no definition of the bonus cost.

Proposal
- It cannot be checked without the definition of the bonus cost.

Evidence Annex B.

24. NO DEFINITION **`prior_churn_score`, `prior_engagement_score`, `prior_risk_score`, `prior_betting_aggression_score`**

Error
- **They are not the value of 7 days before:** they match in 49% to 93% of the rows.
- **The `prior_*` of amounts and counts are** exactly the window before.

Definition
- **Workbook:** it does not say from which date the previous value is.

Proposal
- It cannot be checked without that date.

Evidence Annex B.

25. NO DEFINITION `bet_z_score`, `session_len_z`

Error
- **With the mean and the deviation per brand** they match in 0%–0.3% of the rows.
- **The scores built on them** (`risk_appetite_score`, `betting_aggression_score`) do follow their formula.

Definition
- **Workbook:** $z = (x - \mu)/\sigma$ "of the tenant", without saying which players.

Proposal
- It cannot be checked without knowing which players and which group are used.

Evidence Annex B.

8 Impossible and Suspicious Values, by Brand and Day

Why it matters. An *impossible* value (a share above 1, a session of more than a day) cannot come from real data: it is direct evidence that something is calculated wrong. A *suspicious* value is possible but very rare (more than 50 deposits in a day) and needs a check. Looking at them by brand and day shows where the problem comes from:

- **In every brand and every day** ⇒ it is a calculation error.
- **Only in one brand or on some days** ⇒ it is a specific problem (some data, a release, some players) or real behaviour to check.

Table 1: Impossible and suspicious values: rows, % in all brands and in each brand, days with any, and first and last day; the darker the cell, the higher the % (from 0.1%, 1%, 5%, 10% and 20%). Columns of the signals table: 1 June to 15 October 2026; other metrics: September 2026.

				% of rows						Days		
Variable	Type	Rule	Rows	All	14	23	64	72	73	N	Period	Fig.
`tp_hourly_entropy`	Impossible	above 1	7 189 973	27.77	16.37	38.07	28.77	22.75	29.87	102	06/07–15/10	26
`tp_weekend_index`	Impossible	above 1	6 053 177	23.38	17.16	26.49	26.31	22.48	23.79	102	06/07–15/10	25
`tp_night_index`	Impossible	above 1	2 094 223	8.09	8.50	6.64	9.08	9.63	7.28	102	06/07–15/10	12
Mean session length	Impossible	over 1 day	165 342	1.06	0.96	0.75	0.63	0.57	1.93	98	06/07–15/10	6
Gap between sessions	Impossible	negative	80 390	0.86	1.03	0.31	0.56	1.00	1.30	102	06/07–15/10	28
Median session length	Impossible	over 1 day	116 508	0.75	0.75	0.46	0.40	0.37	1.42	92	06/07–15/10	27
`hold_pct_l30d`	Impossible	over 100%	232	0.00	0.00	0.00	0.00	0.01	0.00	62	17/06–17/08	24
Mean session length	Suspicious	1 s, or 4 to 24 h	1 360 735	8.76	5.46	6.90	6.23	7.50	13.80	102	06/07–15/10	29
Median session length	Suspicious	1 s, or 4 to 24 h	957 103	6.16	3.36	3.13	5.88	8.05	8.29	102	06/07–15/10	30
Gap between sessions	Suspicious	under 30 min	23 335	0.25	0.17	0.15	0.17	0.34	0.34	102	06/07–15/10	31
Bets per day	Suspicious	over 20 000	111	0.02	0.02	0.03	0.00	0.02	0.01	30	01/09–30/09	35
`hold_pct_l30d`	Suspicious	below −1000%	2 469	0.02	0.00	0.01	0.01	0.02	0.04	137	01/06–15/10	32
Deposits per day	Suspicious	over 50	43	0.01	0.00	0.00	0.00	0.02	0.03	25	01/09–30/09	33
Turnover per day	Suspicious	over 1 M€	25	0.00	0.00	0.01	0.00	0.00	0.00	22	01/09–30/09	37
Mean deposit	Suspicious	over 100 000 €	9	0.00	0.00	0.00	0.00	0.00	0.01	9	12/09–29/09	34
Stake in 30 days	Suspicious	over 10 M€	257	0.00	0.00	0.00	0.00	0.00	0.00	136	01/06–15/10	39
Mean bet	Suspicious	over 10 000 €	25	0.00	0.00	0.00	0.00	0.00	0.00	22	19/06–24/09	38
Sessions per day	Suspicious	over 20	1	0.00	0.00	0.00	0.00	0.00	0.00	1	13/09–13/09	36

What they show (the maps per day and brand are in the annex, figures 25 to 39):

- **Time of play: a calculation error.** Night, weekend and entropy above 1, sessions over a day and negative gaps appear **in all five brands and on all 102 days** since 6 July 2026. It does not depend on the brand or the day: the formula is wrong.
- **Suspicious sessions (1 s, or 4 to 24 h): also in every brand.** Between 3% and 14% of the rows, every day; brand 73 is the worst.
- **`hold_pct_l30d` above 100%: a specific problem.** Only 232 rows, from 17 June to 17 August 2026, almost all in brand 72; none after that. It points to some data with negative wins that no longer appear.
- **Extreme deposits and bets: a few players to check.** More than 50 deposits in a day only in brands 72 and 73 (43 cases); mean deposits over 100 000 € only in brand 73 (9 cases, in EUR); more than 20 000 bets in a day in every brand (111 cases, mostly in lira), typical of autoplay or bots. These are single cases, not a calculation error, but the data team should look at them.

A Annex: the Evidence

The figures are grouped like the list. Each figure starts with its finding (bold) and how to read it: what each axis is and, when needed, the formula. Its caption says what it proves and links back to the point.

Calculation of each figure. The caption gives the exact formula of what is drawn. $\#\{\ldots\}$: number of players (or rows) that meet the condition; N: all the players (or rows) of the group; b: one bar of the histogram; y_b: the value of that bar. So "100%" means that every player of the group meets the condition.

Colours, the same in every figure. Red line or bar: the signals table. Blue: the value recomputed from each bet, or the correct definition. Grey: a reference. Red zone: impossible values; orange zone: suspicious values. Each brand has the same colour in every figure.

Sessions and time of play

Points 6, 7 and 8. **What I recompute.** With each bet of the raw transactions (`GAME_BET`, not rolled back): a *session* is a run of bets of one player with no gap over 30 minutes; its *length* is the last bet minus the first one; one-bet sessions have no length and are counted apart (in the table they show as 1 second). The mean per player in the month is compared with `tp_avg_session_duration_secs`. **What to look at.** The table (red) has a tail of sessions over 1 day (red zone, impossible) and of 4 to 24 h (orange zone, suspicious) that the recomputed version (blue) does not have; the centre of the two curves is the same.

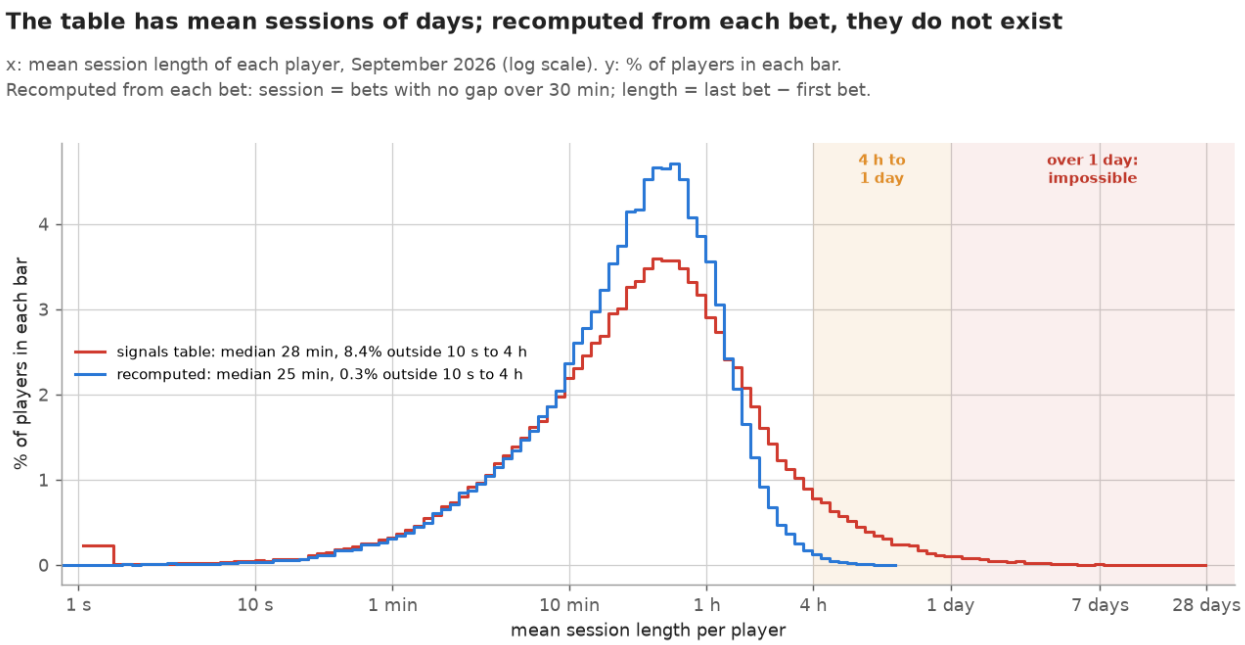


Figure 1: Mean session length. The centre of the two curves is the same; only the table has the tail over 4 h and over 1 day. Point 6. **Calculation:** $y_b = 100\,\#\{p : \bar{L}_p \in b\}/N$, with $\bar{L}_p = \frac{1}{n_p}\sum_s (b_s - a_s)$ the mean length of the n_p sessions of player p and N the players.

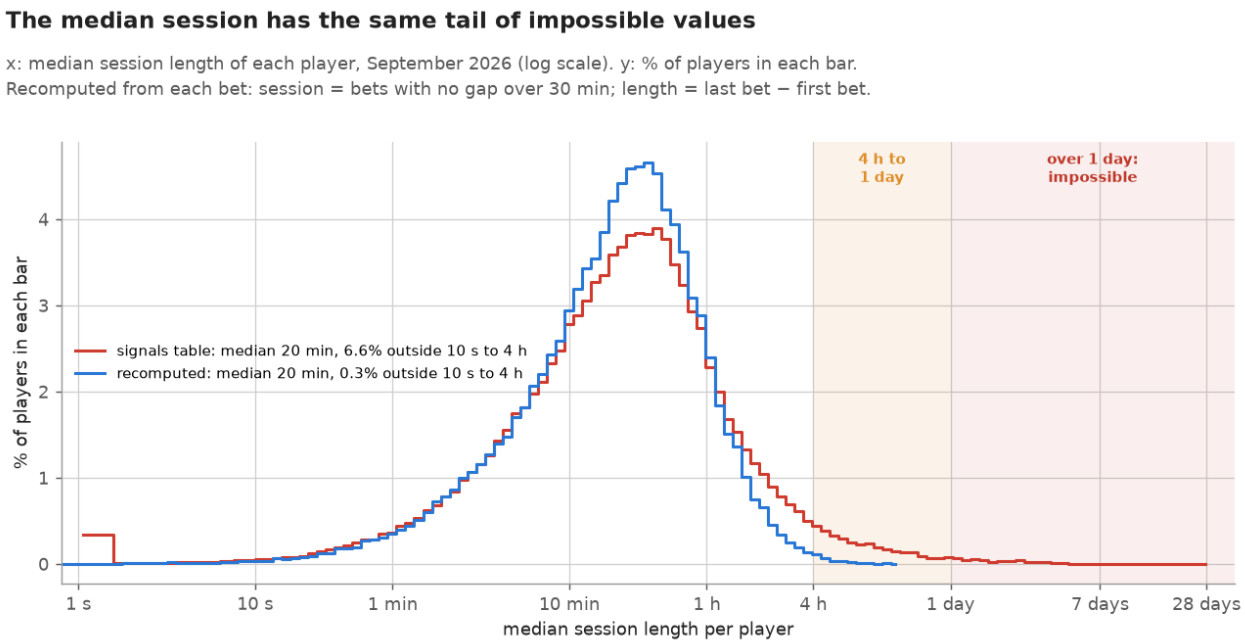


Figure 2: Median session length. The same tail in the median. Point 6. **Calculation:** the same, with the median $\tilde{L}_p = \text{median}_s(b_s - a_s)$.

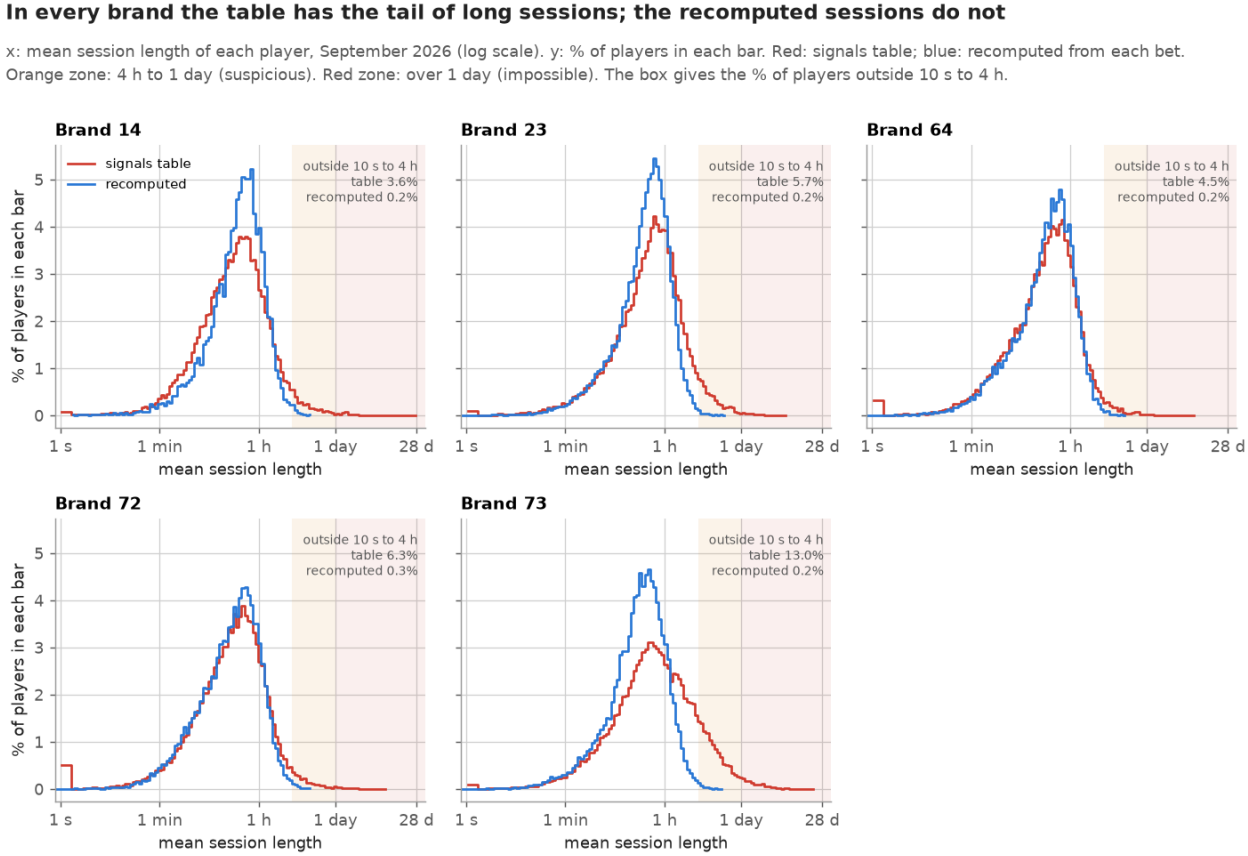


Figure 3: Brand by brand. The tail is in all five brands (3.6% to 13.0% of players outside 10 s to 4 h in the table, 0.2% to 0.3% recomputed); brand 73 is the worst. Point 6. **Calculation:**
$y_b = 100\,\#\{p \in m : \bar{L}_p \in b\}/N_m$ for each brand m; the box is $100\,\#\{p \in m : \bar{L}_p \notin [10\text{ s}, 4\text{ h}]\}/N_m$.

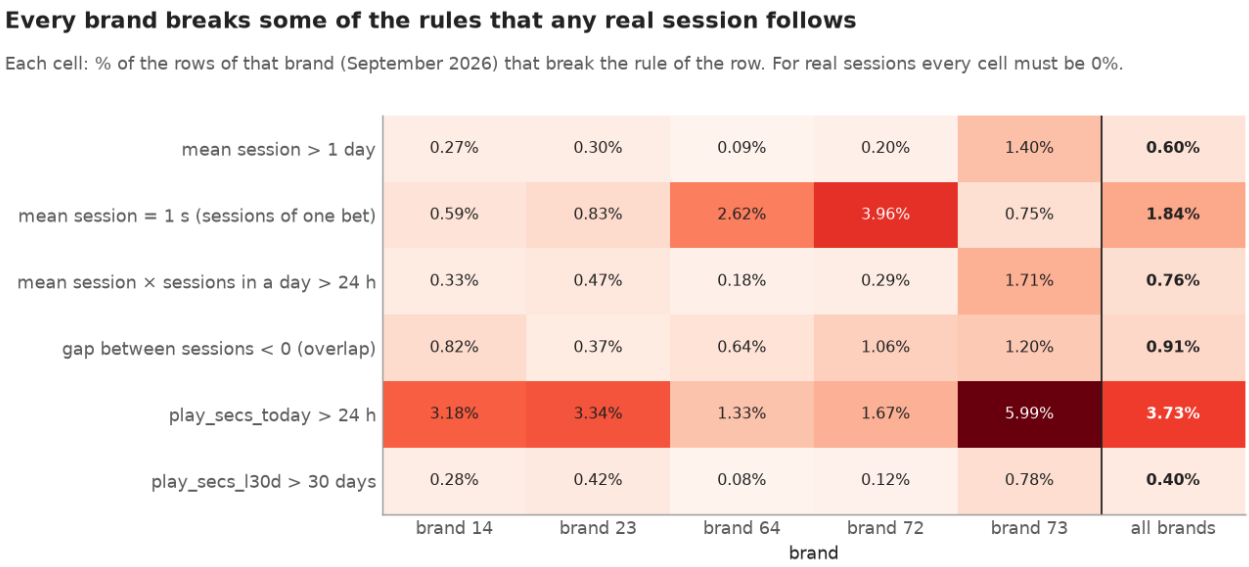


Figure 4: Rules of a real session. Every rule is broken in every brand: the error is in how the table builds the sessions, not in one brand. Points 6, 7 and 8. **Calculation:** each cell $= 100\,\#\{\text{rows of the brand that break the rule}\}/\#\{\text{rows of the brand}\}$.

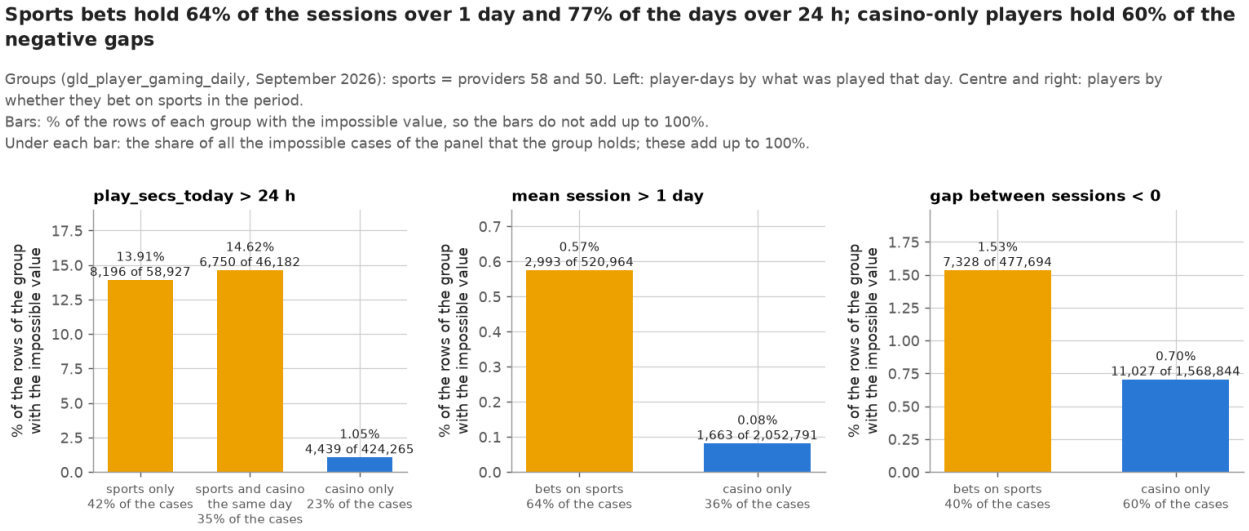


Figure 5: Sports or casino. Most of the time of play over 24 h and of the sessions over 1 day are sports bets; the negative gaps are mostly casino. Points 6, 7 and 8. **Calculation:** for each group G, $100\,\#\{\text{rows of } G \text{ with the impossible value}\}/\#\{\text{rows of } G\}$; below, $100\,\#\{\text{cases of } G\}/\#\{\text{cases}\}$. Sports: providers 58 and 50 of `gld_player_gaming_daily`.

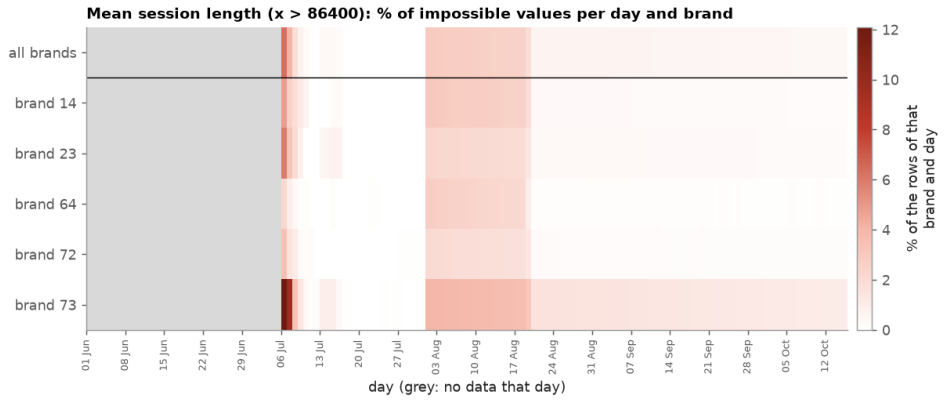


Figure 6: When and where. Sessions over 1 day per day and brand, June to October 2026: in every brand, worst from 1 to 18 August. Point 6.

Night, weekend and entropy

Points 9 to 12. In the workbook the three columns are shares, so they must be between 0 and 1: anything above 1 (red zone) is impossible. Also, the table values fall on exact fractions (dotted lines), which is the sign of an index against an even spread ($3.5/k$, $4/k$) or of an entropy in bits ($\log_2 k$).

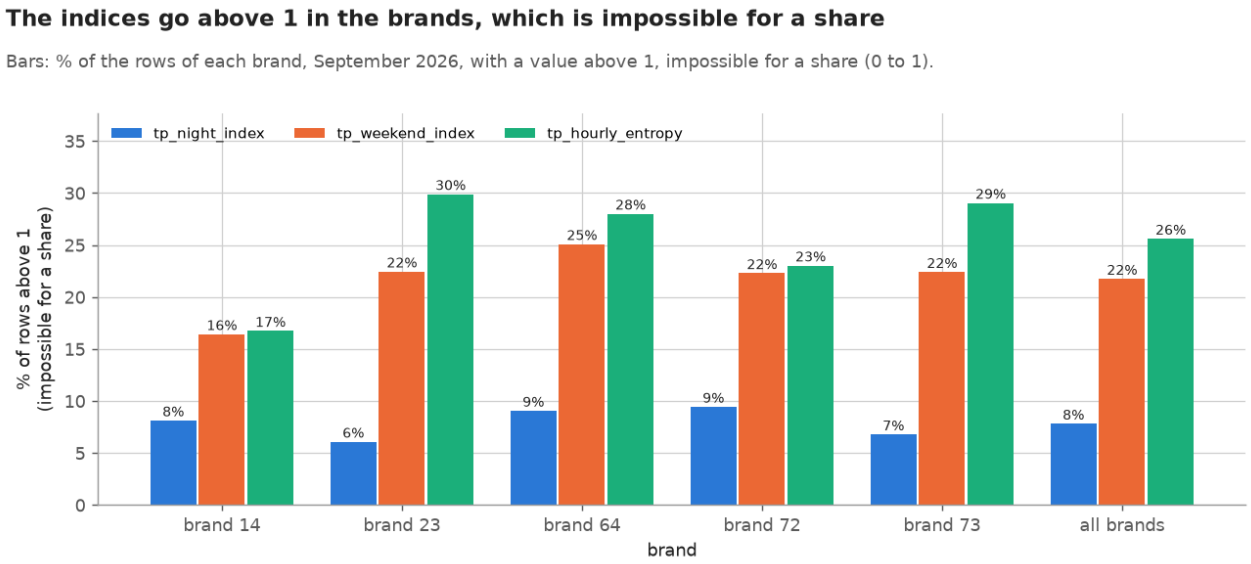


Figure 7: Brand by brand. The same error in every brand. Points 9, 11 and 12. **Calculation:** $100\,\#\{\text{rows with } x > 1\}/N$, with x the value of the column and N the rows of the brand.

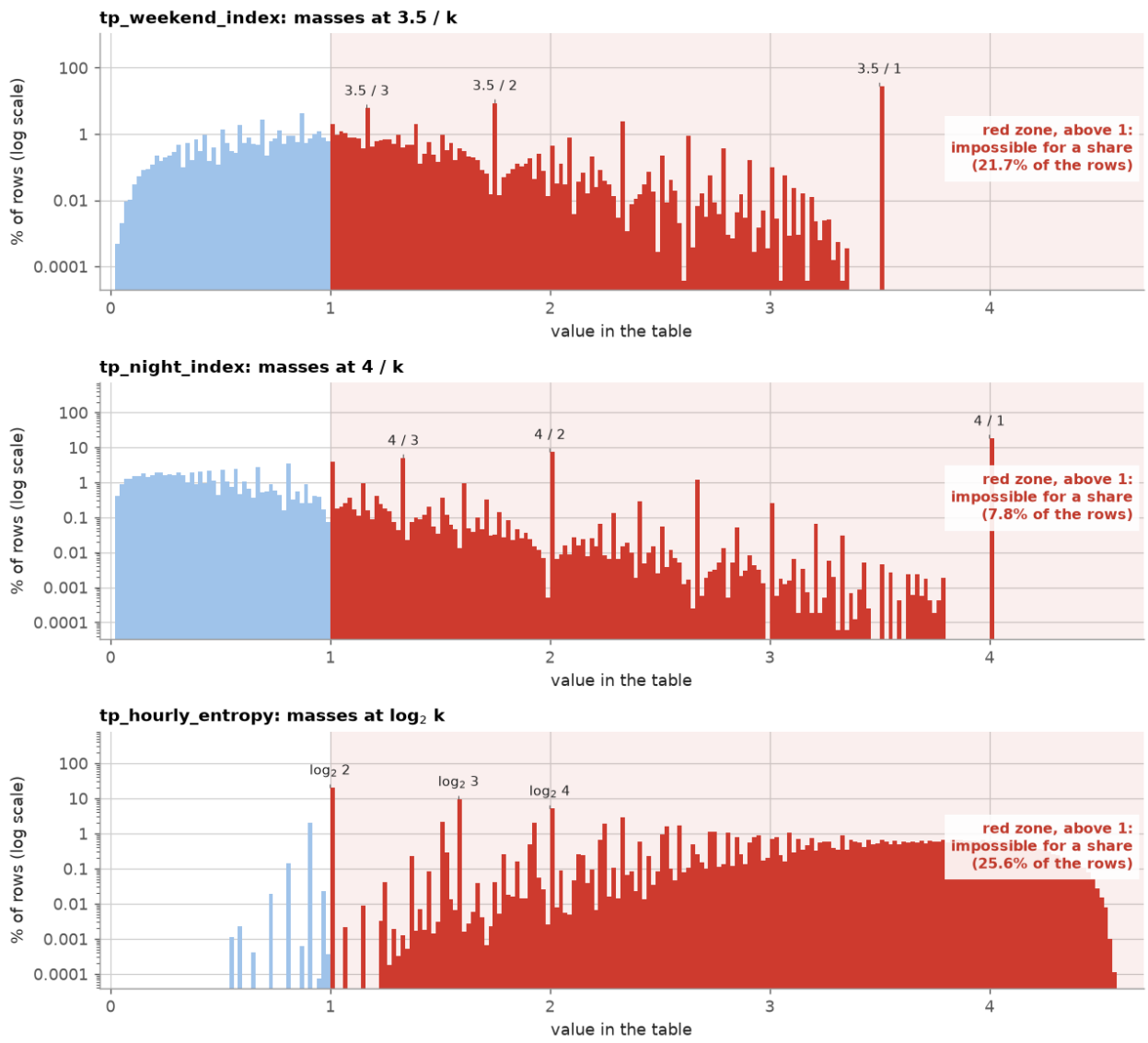


Figure 8: The scale. The masses at $3.5/k$, $4/k$ and $\log_2 k$ show how the table builds each index. Points 9, 11 and 12. **Calculation:** $y_b = 100\,\#\{\text{rows with } x \in b\}/\#\{\text{rows with } x \neq 0\}$.

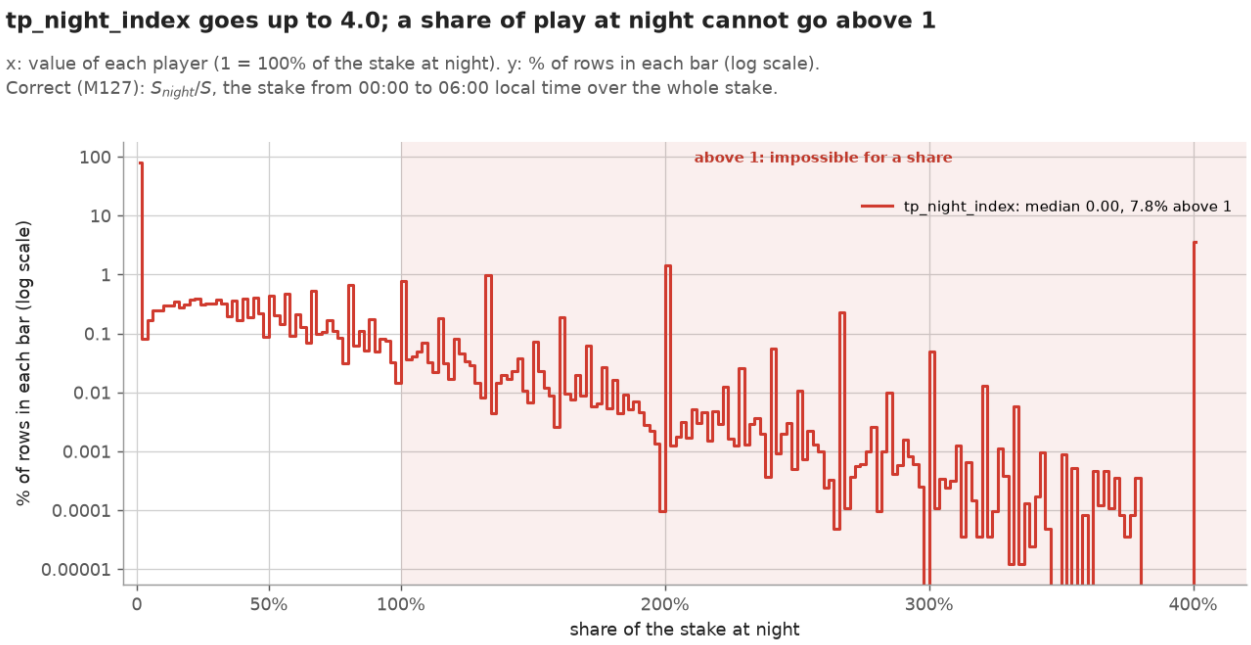


Figure 9: Night. `tp_night_index` goes up to 4. Point 9. **Calculation:** $y_b = 100\,\#\{\text{rows with } x \in b\}/N$.

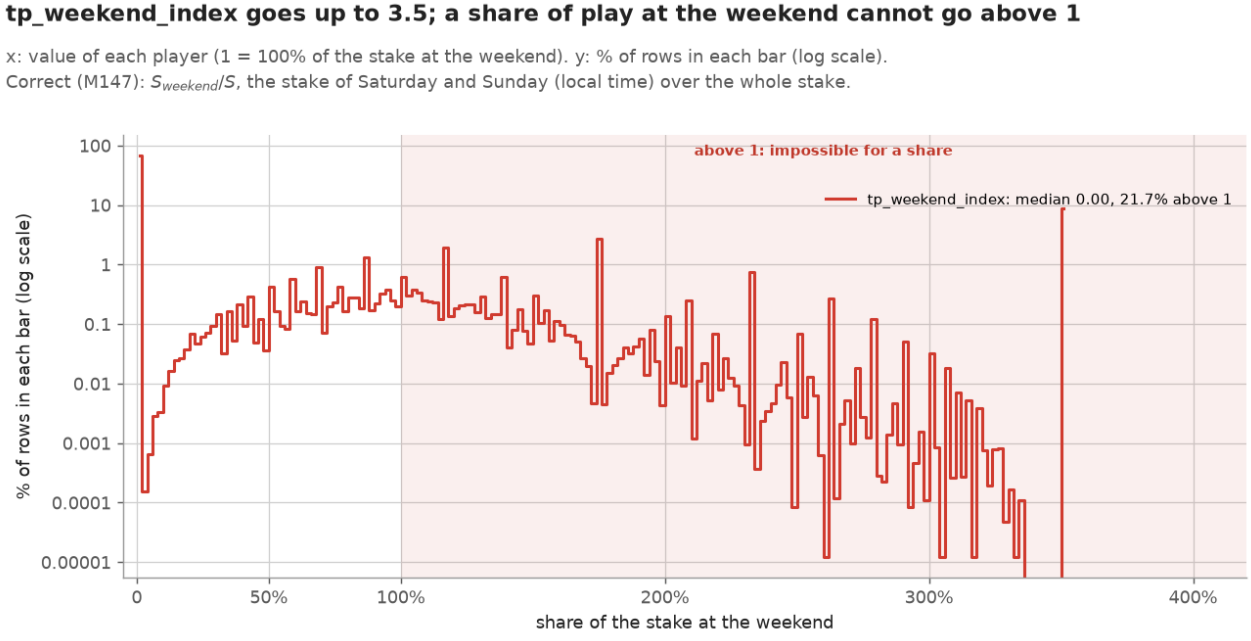


Figure 10: Weekend. Up to 3.5. Point 11. **Calculation:** $y_b = 100\,\#\{\text{rows with } x \in b\}/N$.

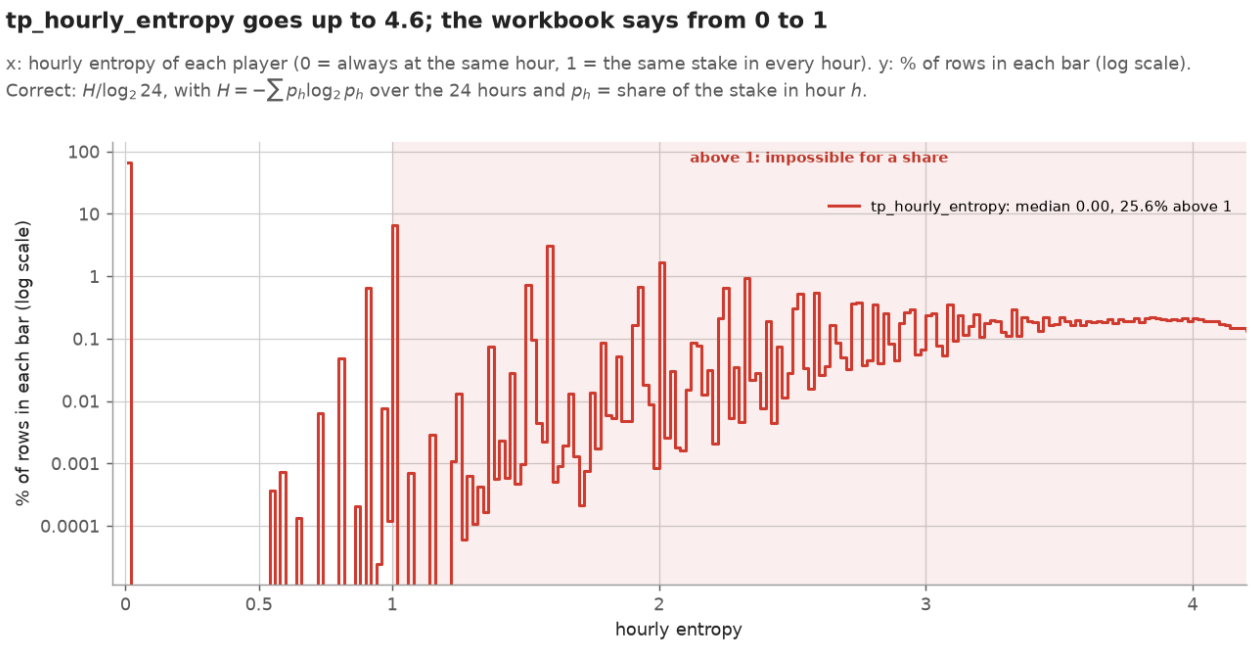


Figure 11: Hourly entropy. Up to 4.6, in bits. Point 12. **Calculation:** $y_b = 100\,\#\{\text{rows with } x \in b\}/N$.

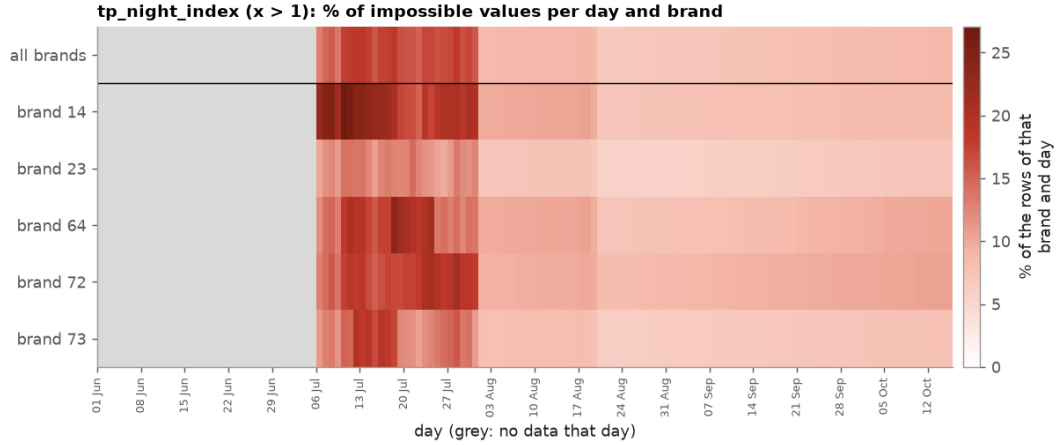


Figure 12: Night above 1, per day and brand. In every brand since 6 July. Points 9 and 5.

Coverage and freshness of time of play

Point 5. A *frozen day* is a day when the time-of-play columns are not updated: they have exactly the same number of rows with data and the same number of impossible values as the day before, to the unit, while the rest of the table (for example, `active_days_l30d`) does change. If they were updated, these numbers would change a little from day to day, as they do on every other day.

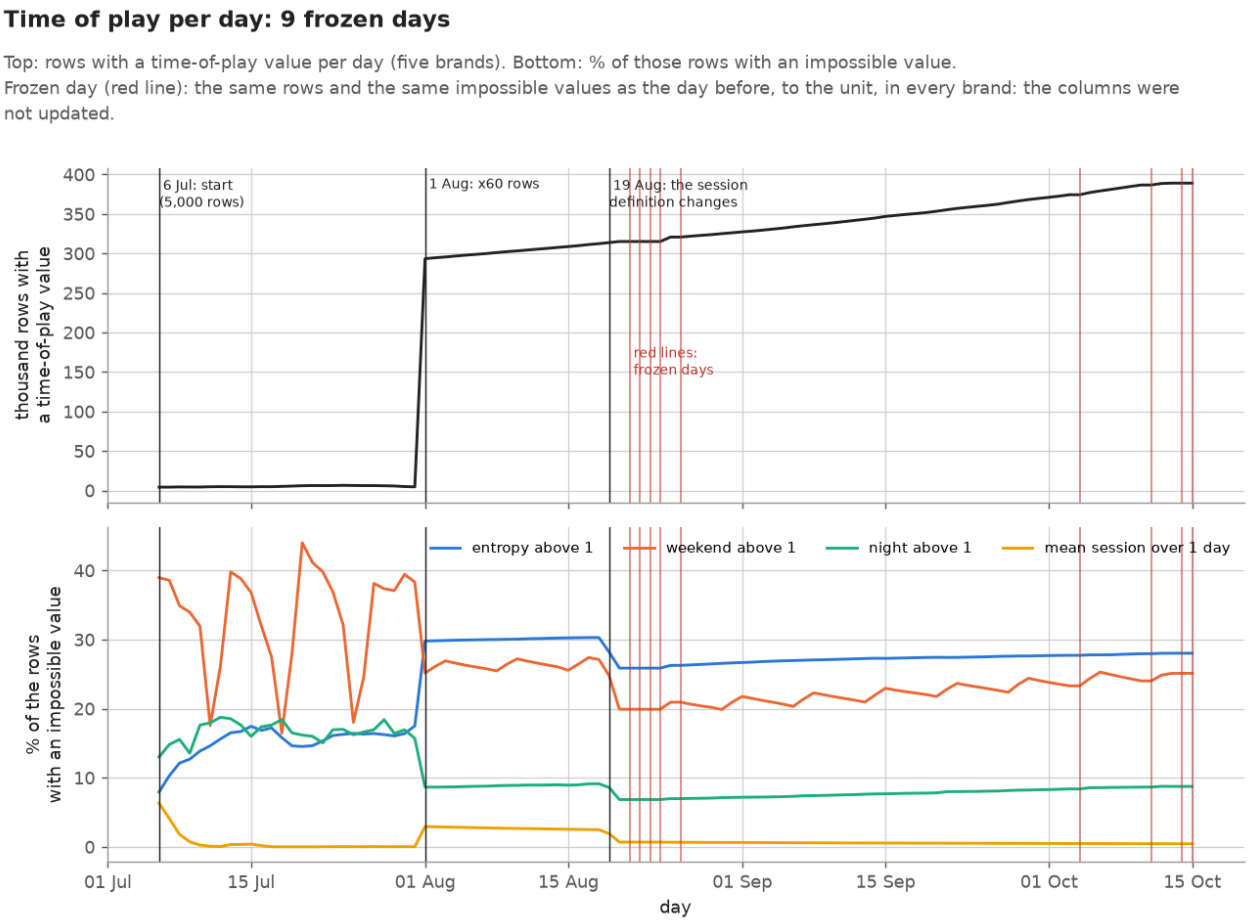


Figure 13: Three stages and frozen days. The history of the `tp_*` columns is not usable, and the daily process needs a freshness check. Point 5. **Calculation:** top, N_d = rows of day d with a value; bottom, $100\,\#\{\text{rows of day } d \text{ with an impossible value}\}/N_d$.

Recency: `days_since_bet`

Points 1 and 3. In each snapshot of the table (date D), I compare `days_since_bet` with the real recency of each player: the days from their last bet up to D, from the daily activity. Empty until May 2026; after that it matches the recency of 7 days later, which is why it predicts the future almost perfectly.

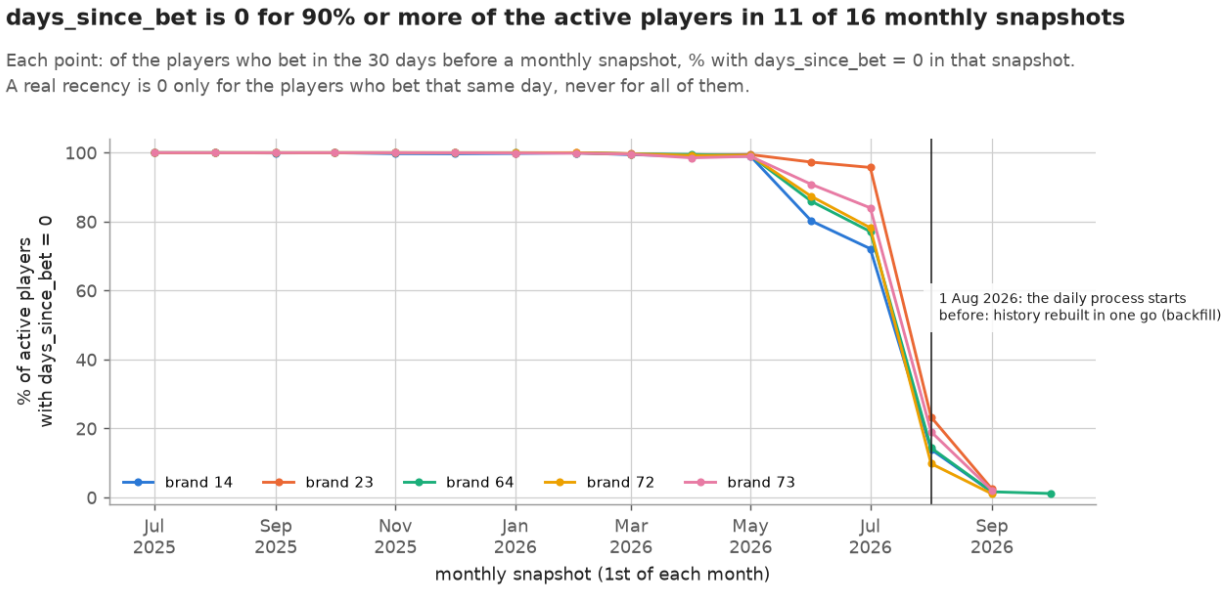


Figure 14: Empty in the history. Before the daily job the column has no information. Point 1. **Calculation:** $100\,\#\{p \in A_D : \text{days_since_bet}_p(D) = 0\}/\#A_D$, with A_D the players with a bet in $(D-30, D]$.

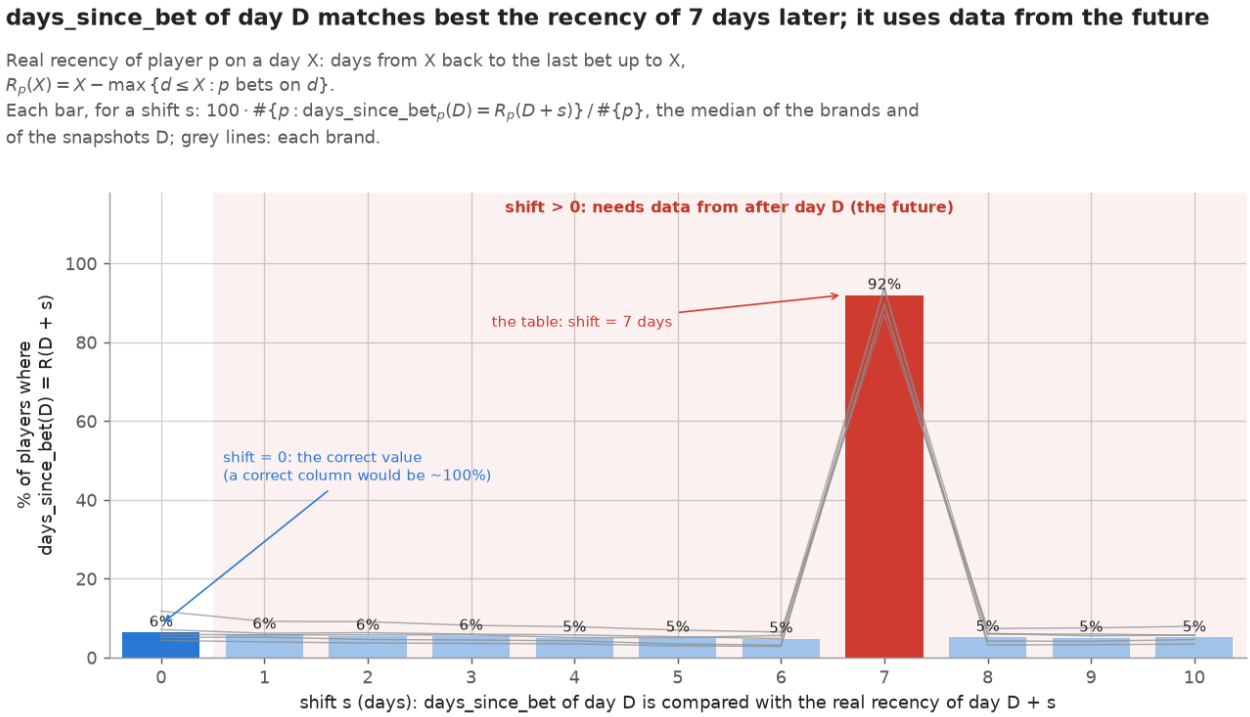


Figure 15: Which day it looks at. For 92% of the players the column is the recency of 7 days later. Point 1. **Calculation:** $y(s) = 100\,\#\{p : \text{days_since_bet}_p(D) = R_p(D+s)\}/N$, with $R_p(X) = X - \max\{d \le X : p \text{ bets on day } d\}$; median of the brands and of the days D.

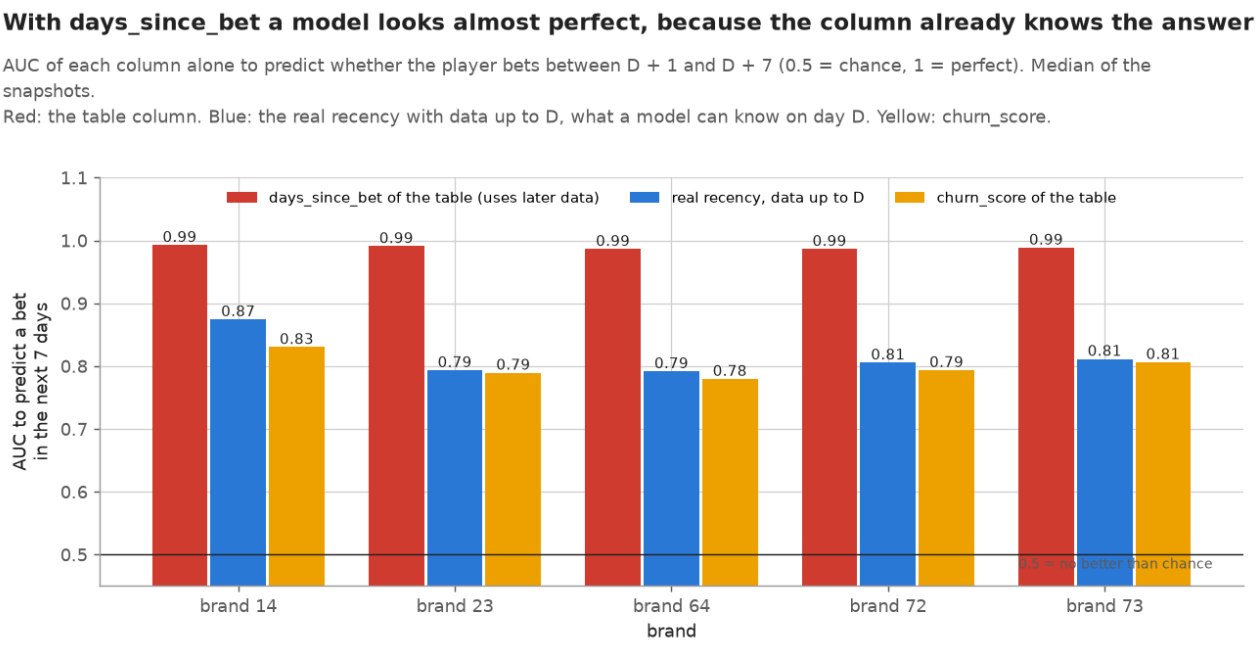


Figure 16: The effect on a model. An AUC of 0.99 is impossible with data up to D. `churn_score` does not show the leak because it mixes other inputs. Points 1 and 3. **Calculation:** AUC of $-x_p$ for the label $y_p = 1$ if p bets in $[D+1, D+7]$, with $x_p =$ `days_since_bet`, $R_p(D)$ or `churn_score`.

Active days and tenure

Points 2 and 14. The table counts the days with any wallet (almost 100% match), not the official active days; tenure does not grow day by day until August 2026.

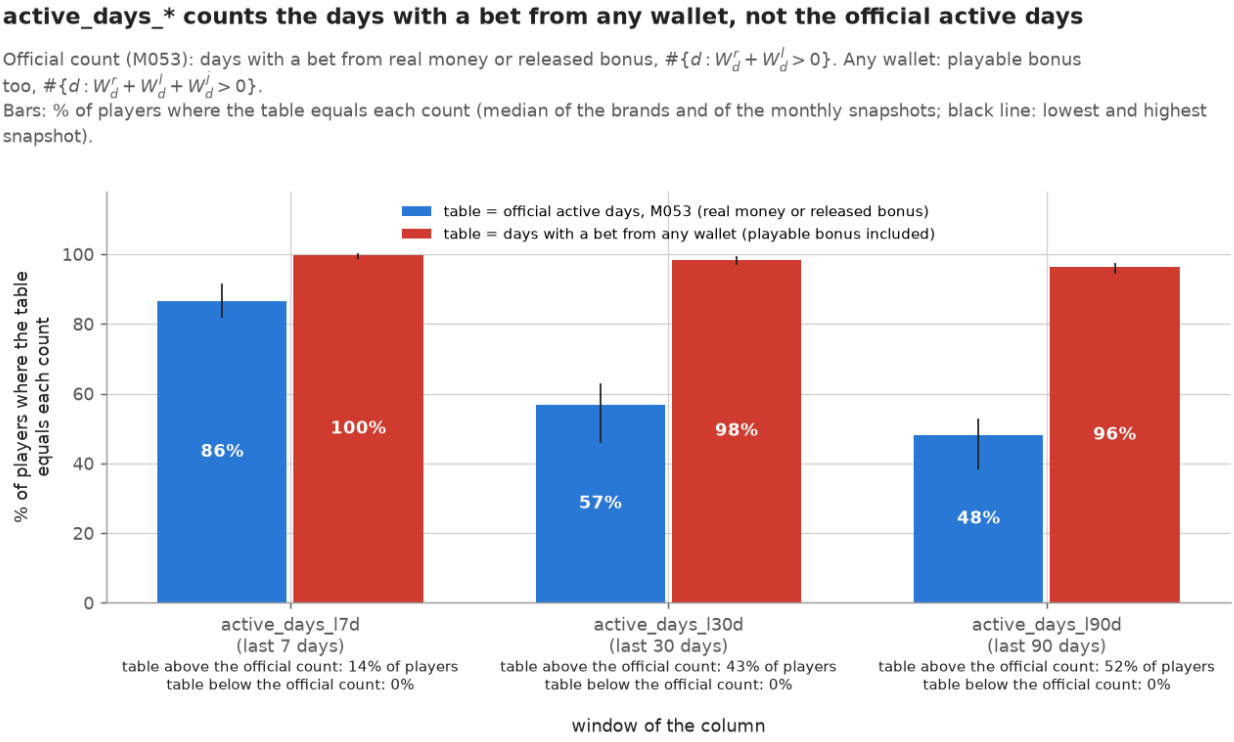


Figure 17: Which definition it uses. The table follows the count with any wallet (96% to 100% of players), not the official one. Point 2. **Calculation:** blue $= 100\,\#\{p : A_p^{\text{table}} = A_p^{\text{official}}\}/N$; red, the same with $A_p^{\text{any wallet}}$; median of the brands and of the snapshots.

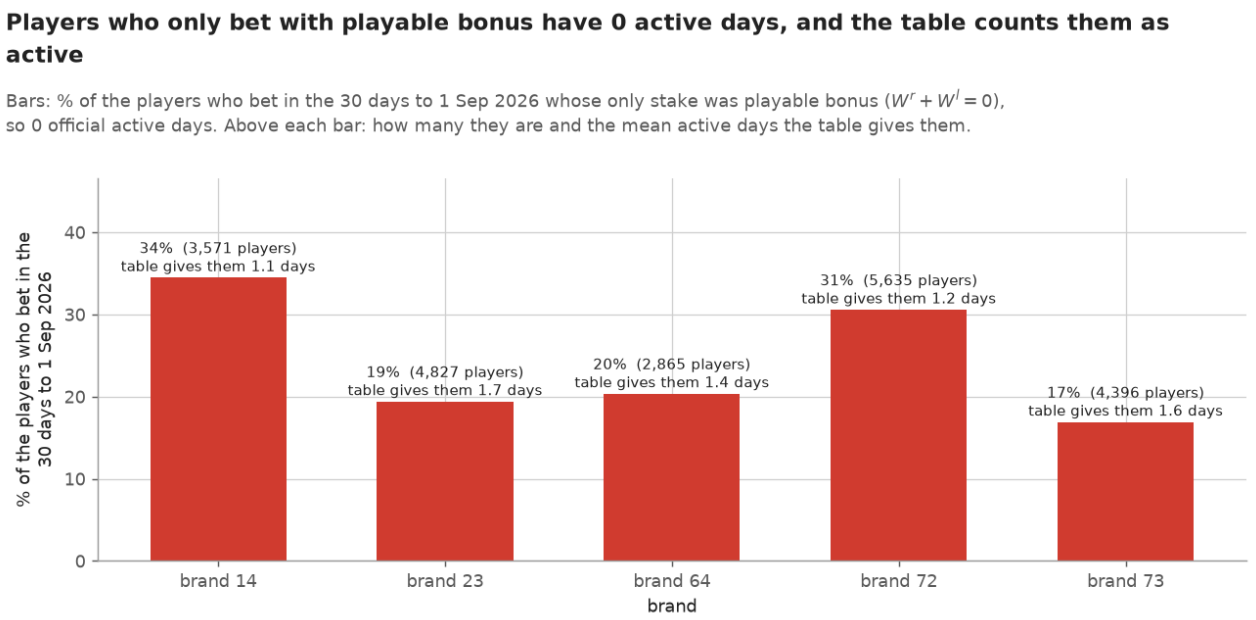


Figure 18: False actives. 17% to 34% of the recent players. Point 2. **Calculation:** $100\,\#\{p : \sum_d (W_d^r + W_d^l) = 0,\ \sum_d W_d^j > 0\}/\#\{p : \text{a bet in the 30 days}\}$.

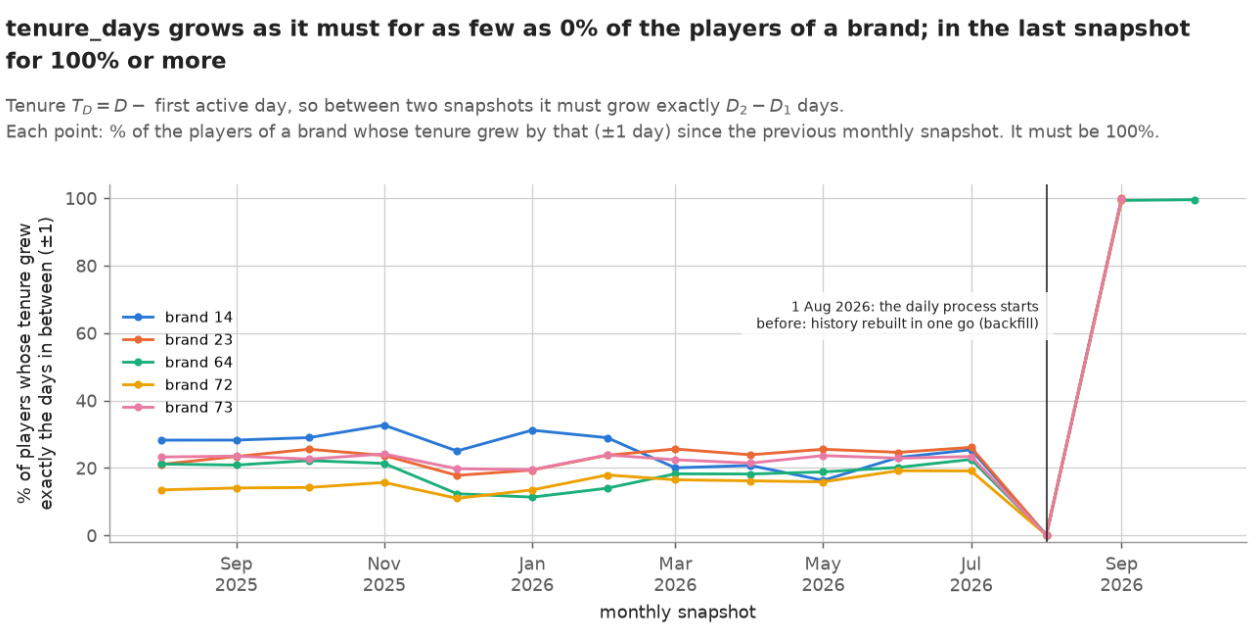


Figure 19: Tenure. Before the daily job it did not grow with time (11% to 33% of the players), on 1 August none grew as it must, and since September it grows correctly (100%). Point 14. **Calculation:** $100\,\#\{p : |T_p(D_2) - T_p(D_1) - (D_2 - D_1)| \le 1\}/N$ for each brand and each pair of consecutive monthly snapshots D_1, D_2.

Scores and indices

Points 4, 15, 18 and 21. Distribution of each column between its limits, per brand; the red bands are the values outside them.

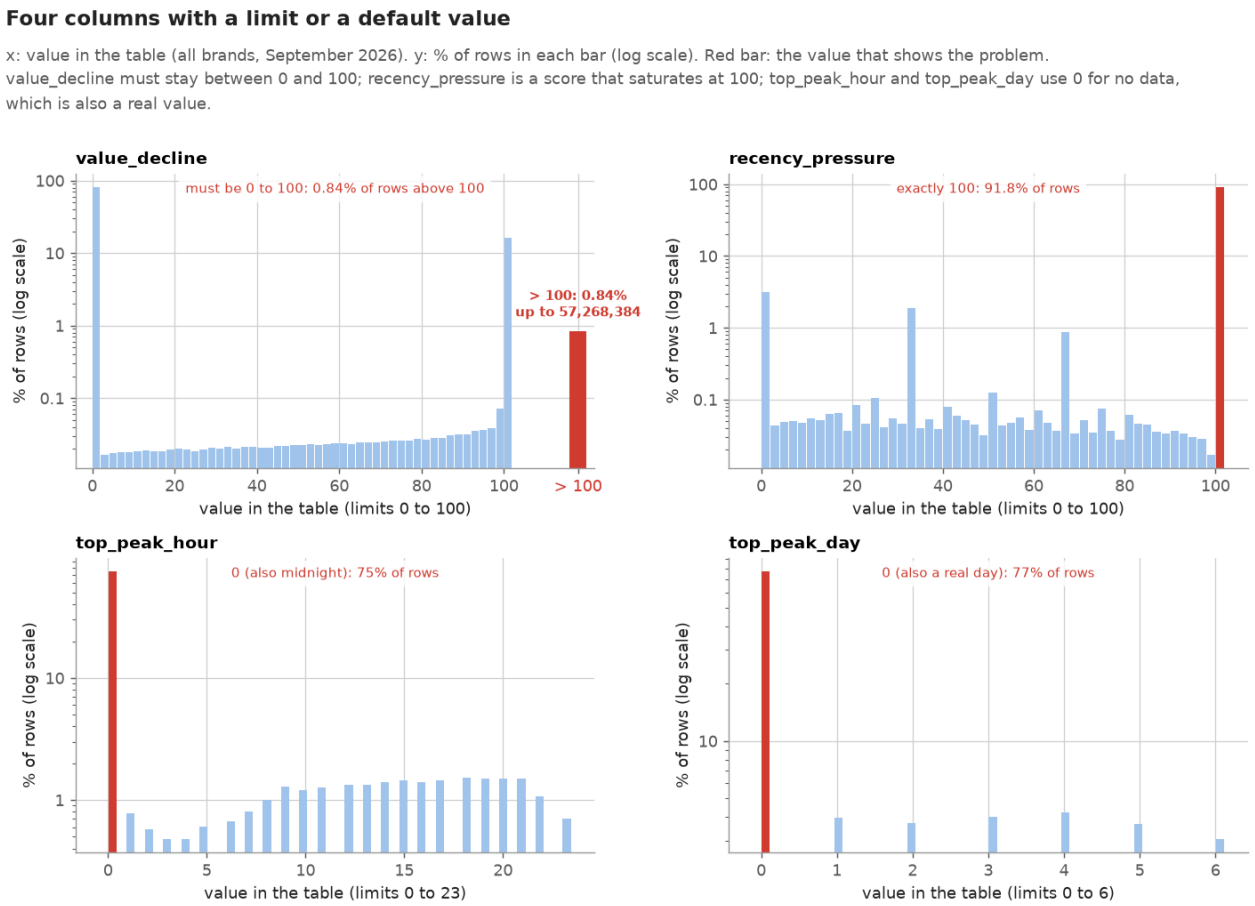


Figure 20: Limits and default values. Points 4, 15, 18 and 21. **Calculation:** $y_b = 100\,\#\{\text{rows with } x \in b\}/N$.

`hold_pct_l30d`

Point 19. Above 100% in the history, with negative wins; the mass at 100% is players with few bets, and the bar at −100% adds up every value below −100%.

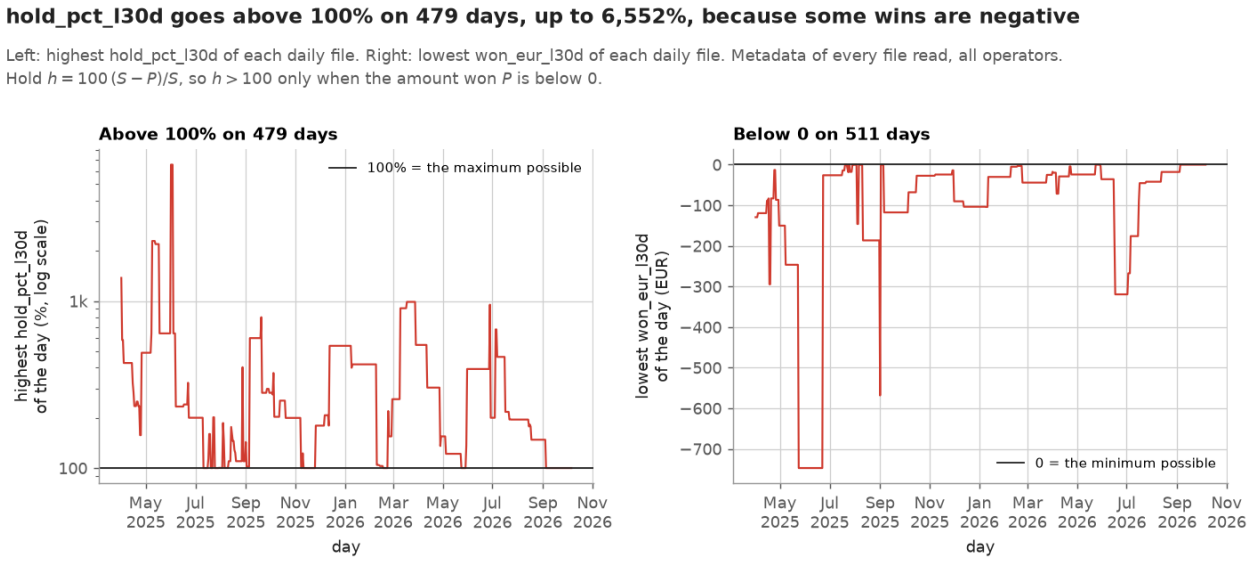


Figure 21: History. The days with a hold above 100% are the days with negative wins. Point 19. **Calculation:** left, $\max_p h_p$ of each daily file; right, $\min_p P_p$.

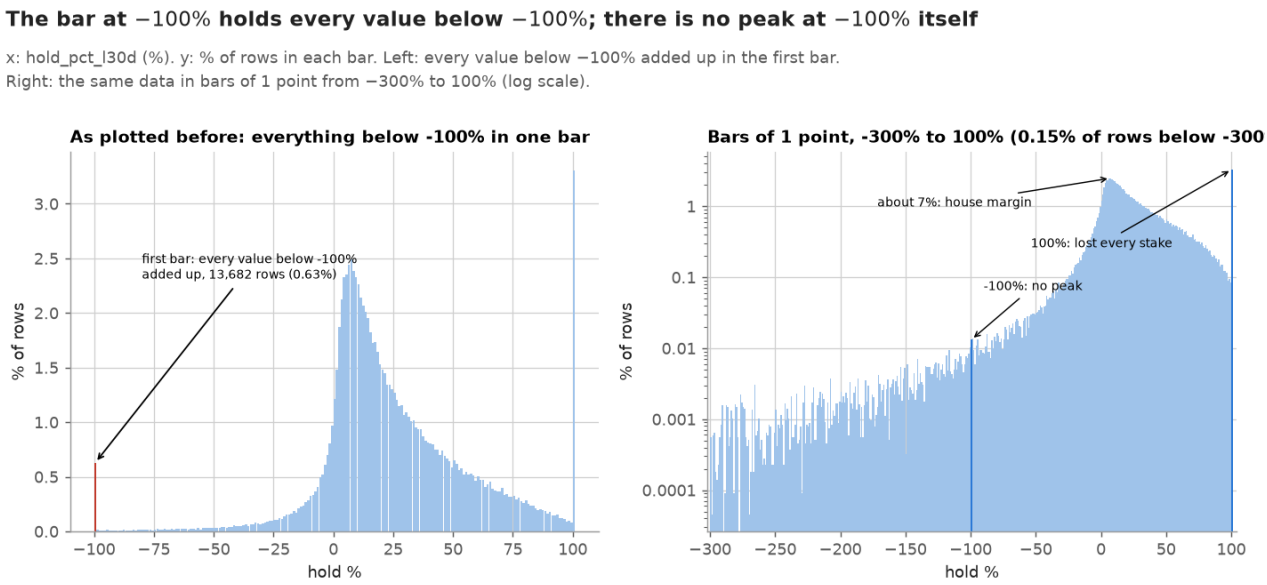


Figure 22: The bar at −100%. It adds up every value below −100%; spread in bars of one point, there is no peak at −100%. Point 19. **Calculation:** $y_b = 100\,\#\{\text{rows with } h \in b\}/N$.

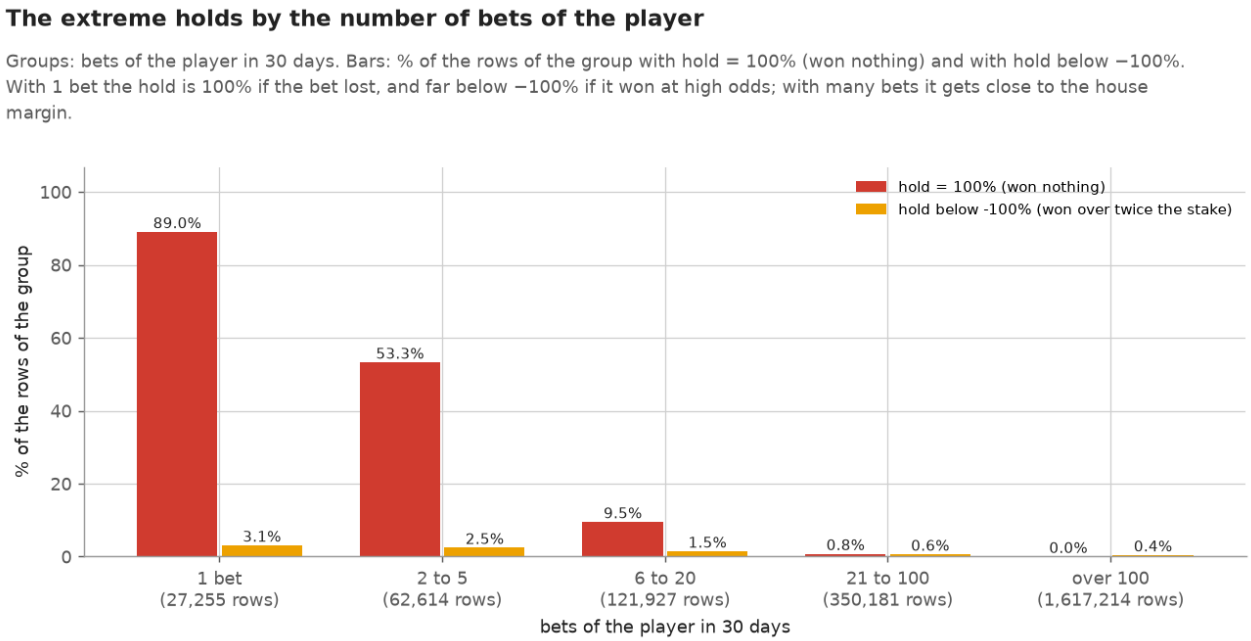


Figure 23: By number of bets. The extreme holds come from players with few bets. Point 19. **Calculation:** $100\,\#\{\text{rows of the group with } h = 100\}/\#\{\text{rows of the group}\}$, and the same with $h < -100$.

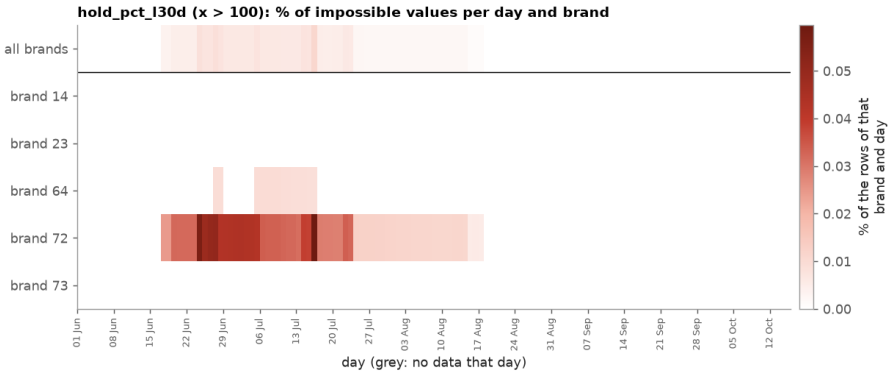


Figure 24: Above 100%, per day and brand. From 17 June to 17 August 2026, almost all in brand 72. Point 19.

Impossible and suspicious values, per day and brand

Section 8. Each map is one variable: rows = brands (the first one, all together), columns = days; the colour is the % of rows of that day and brand that break the rule (white, 0%; grey, no data). A band of colour in all rows and all days is a calculation error; single spots are specific cases.

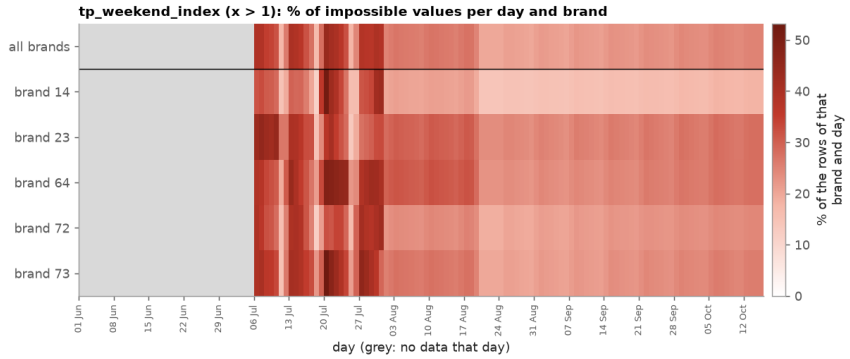


Figure 25: Impossible: `tp_weekend_index` above 1. Section 8.

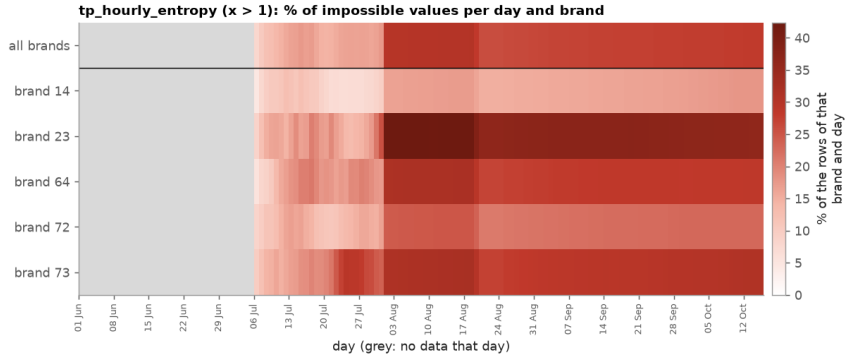


Figure 26: Impossible: `tp_hourly_entropy` above 1. Section 8.

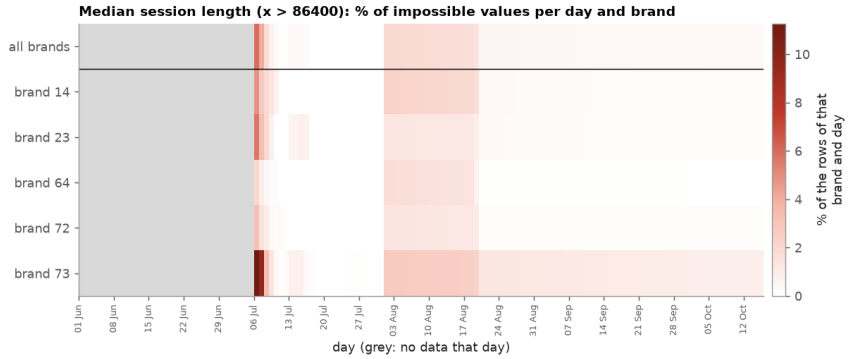


Figure 27: Impossible: median session over 1 day. Section 8.

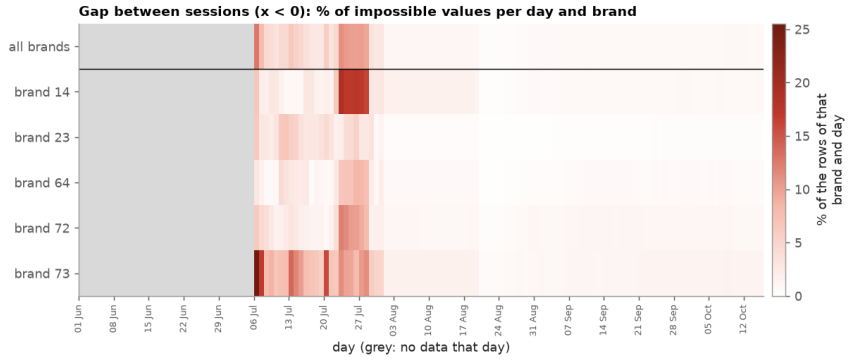


Figure 28: Impossible: negative gap between sessions. Section 8.

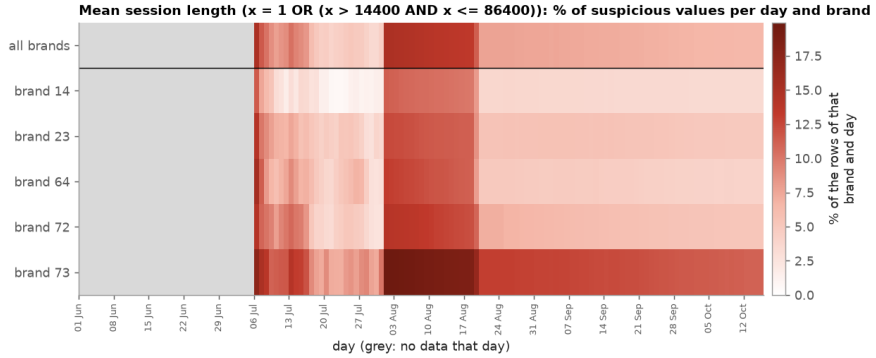


Figure 29: Suspicious: mean session of 1 s or of 4 to 24 h. Section 8.

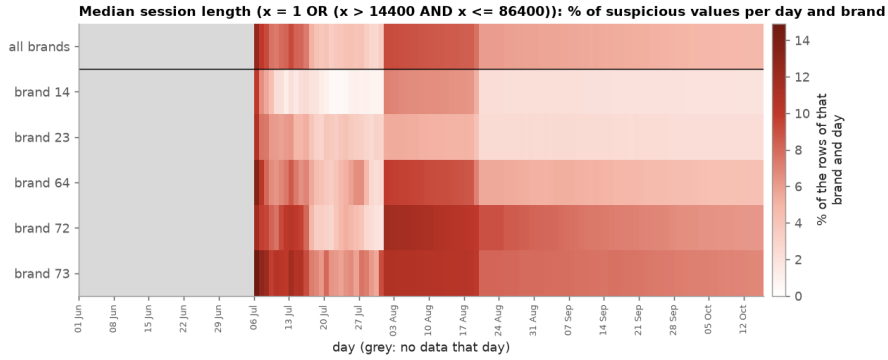


Figure 30: Suspicious: median session of 1 s or of 4 to 24 h. Section 8.

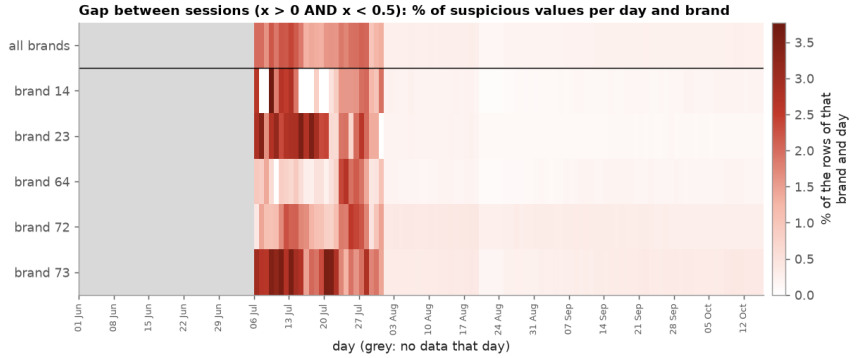


Figure 31: Suspicious: gap between sessions under 30 min. Section 8.

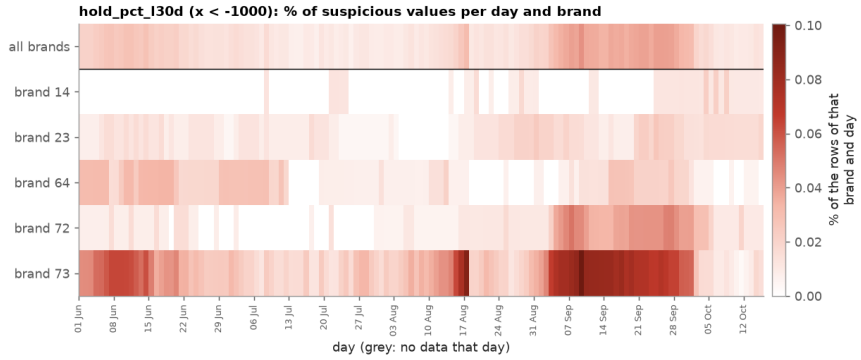


Figure 32: Suspicious: hold below −1000%. Section 8.

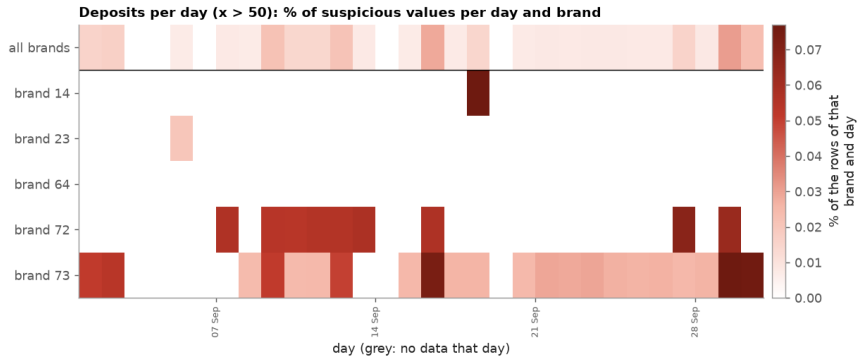


Figure 33: Suspicious: more than 50 deposits in a day. Section 8.

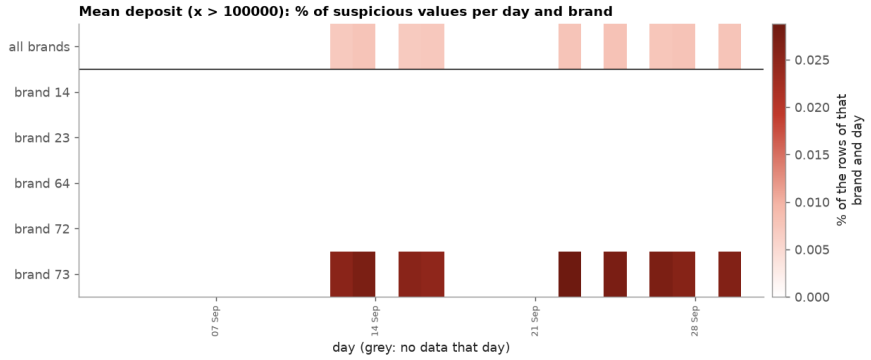


Figure 34: Suspicious: mean deposit over 100 000 €. Section 8.

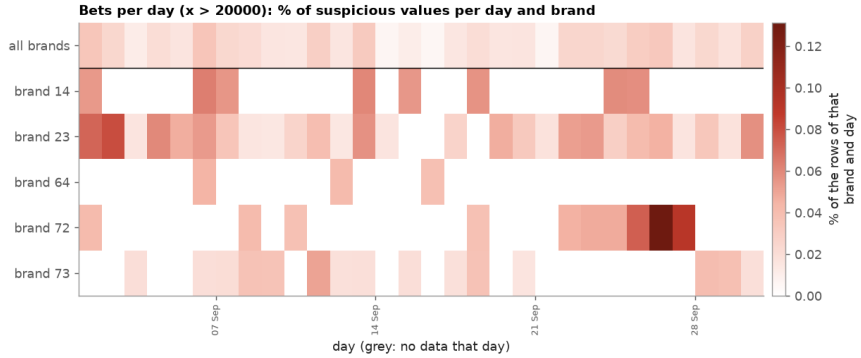


Figure 35: Suspicious: more than 20 000 bets in a day. Section 8.

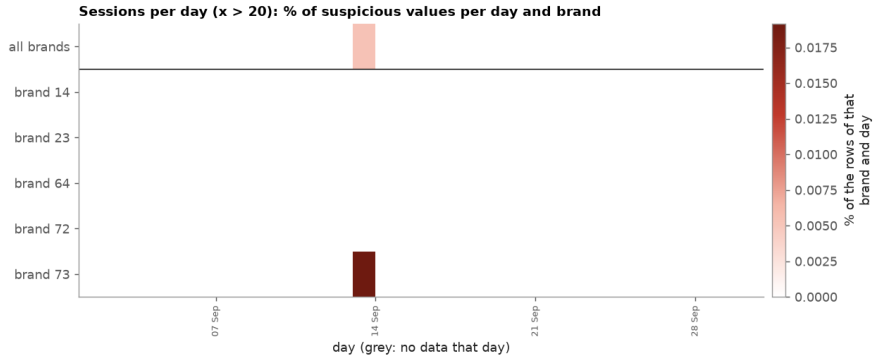


Figure 36: Suspicious: more than 20 sessions in a day. Section 8.

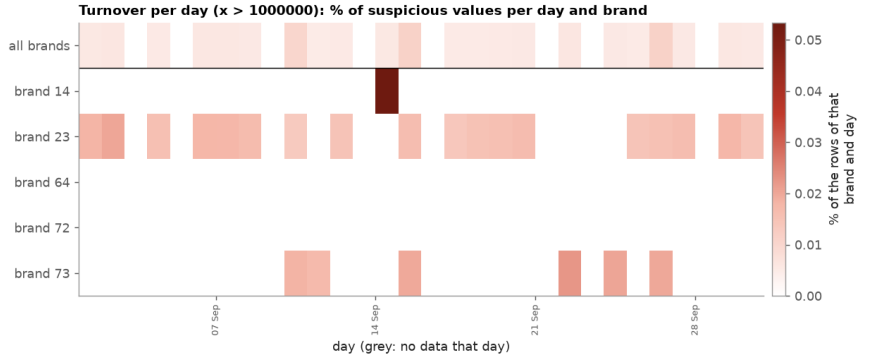


Figure 37: Suspicious: turnover over 1 M€ in a day. Section 8.

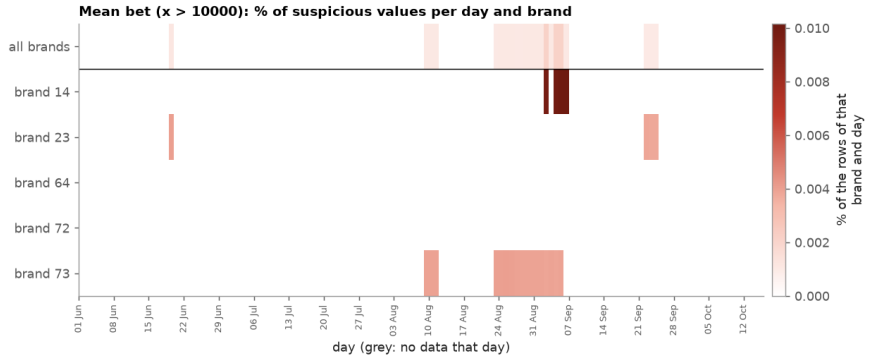


Figure 38: Suspicious: mean bet over 10 000 €. Section 8.

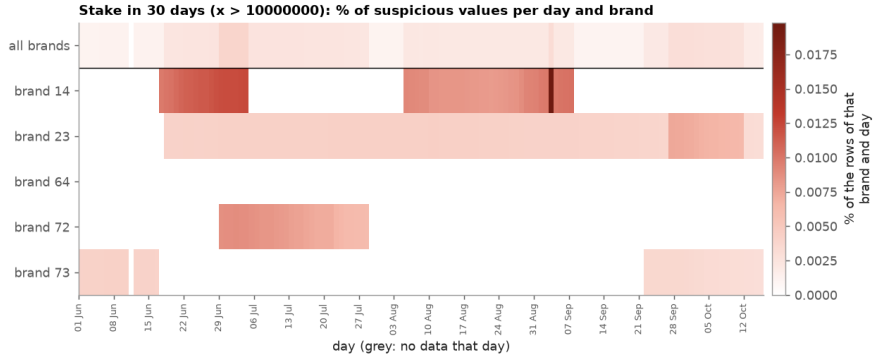


Figure 39: Suspicious: more than 10 M€ staked in 30 days. Section 8.

Keys: `brand_id`

Point 16. Metadata of all daily files, all operators.

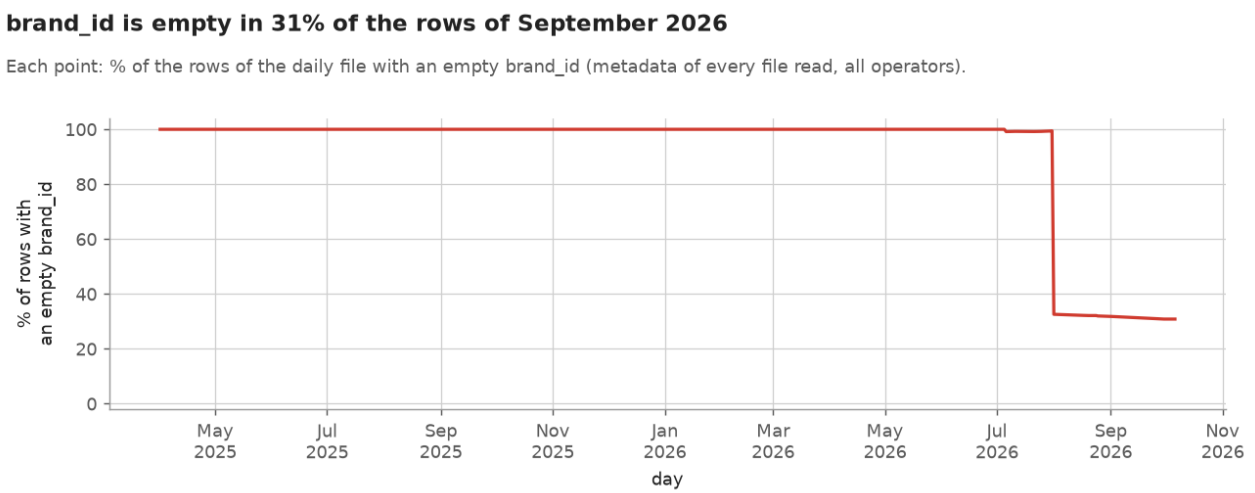


Figure 40: Empty `brand_id`. Point 16. **Calculation:** 100 #{rows with an empty brand_id}/#{rows} of each daily file.

Codes and categories

Point 17 and criterion 2 (distribution). Every day of September 2026, the five brands.

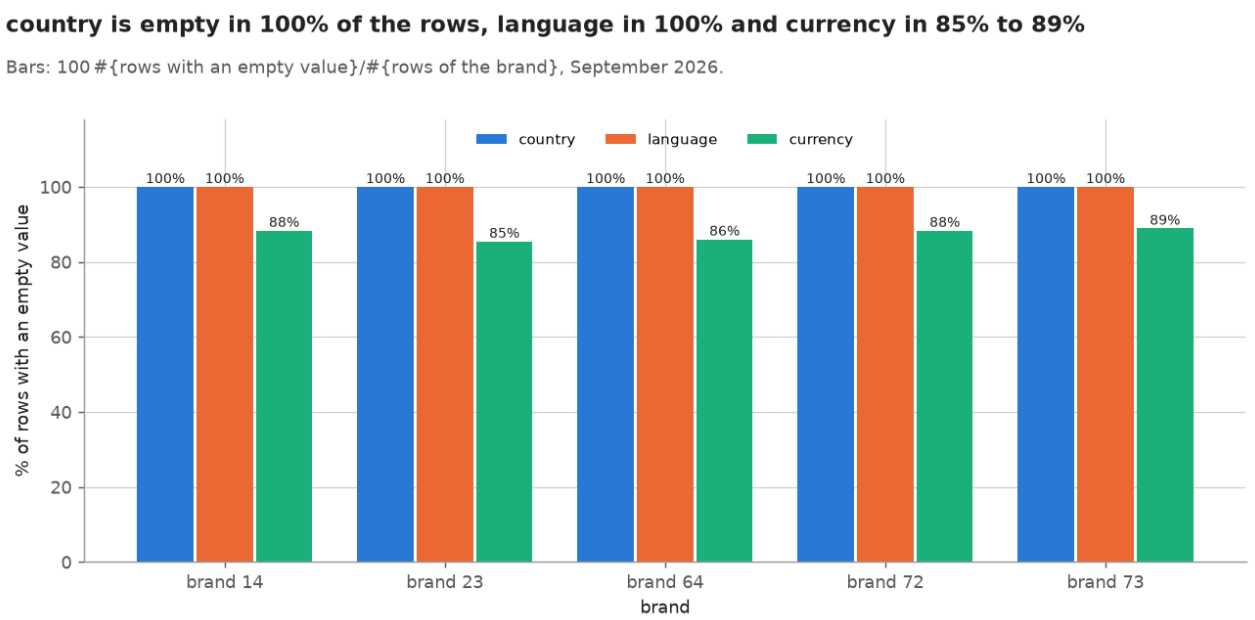


Figure 41: Empty codes. Point 17. **Calculation:** 100 #{rows with an empty value}/N_m per brand m.

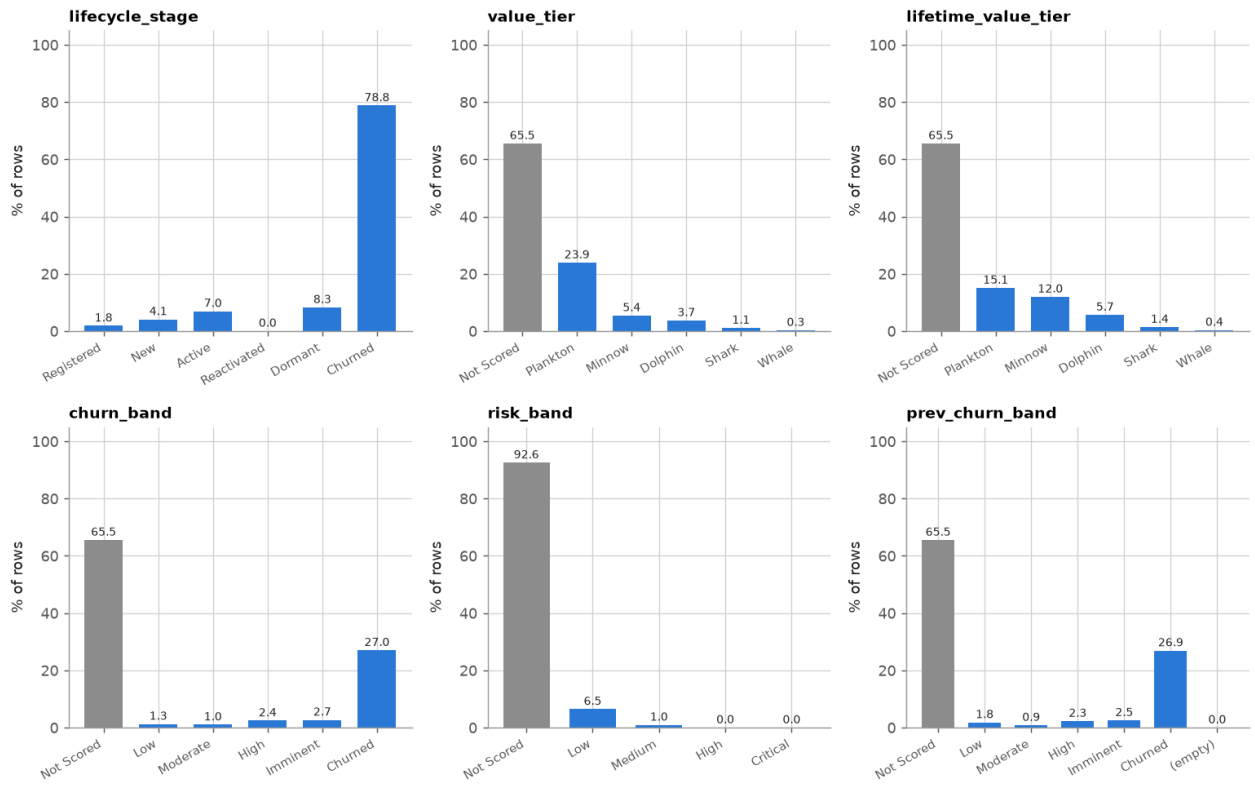


Figure 42: Tiers and bands: right. Not uniform: the value and risk levels fall from the first to the last, as expected. *Not Scored* are players without enough data to compute the score. Criterion 2. **Calculation:** $y_v = 100\,\#\{\text{rows with the value } v\}/N$.

B Annex: the 165 Columns and Their Checks

The 165 columns and their 210 checks, sorted by status and, the wrong ones, by the priority of their point. Each column is on a grey or white band, with its name, its status and, if it is in the list, the priority and number of its point, once. *Rule*: the one that runs, DuckDB SQL on the columns of the table. *Pass*: % of the rows of the two snapshots that meet the rule (criterion 3), empty if it is not calculated here; green if it passes ($\geq$ 99%), orange if almost (95% to 99%) and red if it fails. A wrong column can pass its checks at 100%: the error is in its definition or in an input, and it shows when it is compared with the daily tables.

Column	Status and point	Kind	Rule	Pass (%)
`active_days_l180d`	Wrong HIGH 2	Limit	`active_days_l180d >= active_days_l90d`	100.00
		Limit	`active_days_l180d <= 180`	100.00
`active_days_l30d`	Wrong HIGH 2	Window	= sum of `active_days_today` over the 30 days up to D	99.99
		Limit	`active_days_l30d >= active_days_l7d`	100.00
		Limit	`active_days_l30d <= 30`	100.00
`active_days_l7d`	Wrong HIGH 2	Window	= sum of `active_days_today` over the 7 days up to D	100.00
		Limit	`active_days_l7d >= active_days_today`	100.00
		Limit	`active_days_l7d <= 7`	100.00
`active_days_l90d`	Wrong HIGH 2	Limit	`active_days_l90d >= active_days_l30d`	100.00
		Limit	`active_days_l90d <= 90`	100.00
`active_days_today`	Wrong HIGH 2	Limit	`active_days_today >= 0`	100.00
		Limit	`active_days_today <= 1`	100.00
`days_since_bet`	Wrong HIGH 1	Validated	recalculated from the daily tables	
		Limit	`days_since_bet <= tenure_days`	100.00
`prior_active_days_l30d`	Wrong HIGH 2	Lag	= `active_days_l30d`, 30 days before	100.00
		Limit	`prior_active_days_l30d <= 30`	100.00
`prior_active_days_l7d`	Wrong HIGH 2	Lag	= `active_days_l7d`, 7 days before	100.00
		Limit	`prior_active_days_l7d <= 7`	100.00
`value_decline`	Wrong HIGH 4	Formula	= `CASE WHEN ggr_eur_l90d <> 0 THEN 100 * greatest(0, 1 - (ggr_eur_l7d / 7) / (ggr_eur_l90d / 90)) END`	100.00
`brand_id`	Wrong MEDIUM 16	No check	—	
`country`	Wrong MEDIUM 17	Format	`country <> '' AND country = upper(country)`	0.00

Column	Status and point	Kind	Rule	Pass (%)
currency	Wrong MEDIUM 17	Format	currency <> '' AND currency = upper(currency)	12.43
deposit_recency_score	Wrong MEDIUM 13	Formula	= CASE WHEN deposits_eur_today > 0 THEN 100 ELSE 100 * (1 - least(1, days_since_bet / 30.0)) END	100.00
language	Wrong MEDIUM 17	Format	language <> '' AND language = lower(language)	0.00
play_secs_l180d	Wrong MEDIUM 8	Limit	play_secs_l180d >= play_secs_l90d	100.00
play_secs_l30d	Wrong MEDIUM 8	Window	= sum of play_secs_today over the 30 days up to D	99.95
		Limit	play_secs_l30d >= play_secs_l7d	100.00
play_secs_l7d	Wrong MEDIUM 8	Window	= sum of play_secs_today over the 7 days up to D	99.99
		Limit	play_secs_l7d >= play_secs_today	100.00
play_secs_l90d	Wrong MEDIUM 8	Limit	play_secs_l90d >= play_secs_l30d	100.00
play_secs_today	Wrong MEDIUM 8	Limit	play_secs_today >= 0	100.00
recency_pressure	Wrong MEDIUM 15	Limit	recency_pressure <= least(100, 100 * days_since_bet / 3.0) + 0.01	100.00
rg_night_play_ratio	Wrong MEDIUM 10	Identity	= tp_night_index	91.46
		Limit	rg_night_play_ratio BETWEEN 0 AND 1	100.00
tenure_days	Wrong MEDIUM 14	Validated	recalculated from the daily tables	
tp_avg_session_duration_secs	Wrong MEDIUM 6	Limit	CASE WHEN tp_avg_session_duration_secs > 0 THEN tp_avg_session_duration_secs BETWEEN 10 AND 14400 END	92.70
tp_avg_session_gap_hours	Wrong MEDIUM 7	Limit	tp_avg_session_gap_hours >= 0	99.88
tp_hourly_entropy	Wrong MEDIUM 12	Limit	tp_hourly_entropy BETWEEN 0 AND 1	88.97
tp_median_session_duration_secs	Wrong MEDIUM 6	Limit	CASE WHEN tp_median_session_duration_secs > 0 THEN tp_median_session_duration_secs BETWEEN 10 AND 14400 END	94.56
tp_night_index	Wrong MEDIUM 9	Limit	tp_night_index BETWEEN 0 AND 1	96.82
tp_weekend_index	Wrong MEDIUM 11	Limit	tp_weekend_index BETWEEN 0 AND 1	90.64

Column	Status and point	Kind	Rule	Pass (%)
deposit_frequency_score	Wrong LOW 20	Formula	= CASE WHEN deposits_eur_l30d > 0 THEN least(100, greatest(0, 50 + 50 * least(1, greatest(-1, (deposits_eur_l7d / 7) / (deposits_eur_l30d / 30) - 1)))) ELSE 0 END	100.00
hold_pct_l30d	Wrong LOW 19	Identity	= CASE WHEN wagered_eur_l30d > 0 THEN 100 * ggr_eur_l30d / wagered_eur_l30d END	100.00
		Limit	hold_pct_l30d <= 100	100.00
tax_eur_l180d	Wrong LOW 22	Share	tax_eur_l180d > 0 AND tax_eur_l180d <= ggr_eur_l180d	0.00
tax_eur_l30d	Wrong LOW 22	Window	= sum of tax_eur_today over the 30 days up to D	100.00
tax_eur_l7d	Wrong LOW 22	Window	= sum of tax_eur_today over the 7 days up to D	100.00
tax_eur_l90d	Wrong LOW 22	Share	tax_eur_l90d > 0 AND tax_eur_l90d <= ggr_eur_l90d	0.00
tax_eur_today	Wrong LOW 22	Share	tax_eur_today > 0 AND tax_eur_today <= ggr_eur_today	0.00
top_peak_day	Wrong LOW 18	Limit	top_peak_day BETWEEN 0 AND 7	100.00
top_peak_hour	Wrong LOW 18	Limit	top_peak_hour BETWEEN 0 AND 23	100.00
churn_band	Affected HIGH 3	Band	bands of churn_score, cutoffs 25, 50, 75	
		Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
churn_propensity_raw	Affected HIGH 3	Formula	= least(1, days_since_bet / 90.0)	100.00
churn_score	Affected HIGH 3	Formula	= least(100, greatest(0, 0.50 * recency_pressure + 0.30 * frequency_decline + 0.20 * value_decline))	100.00
engagement_score	Affected HIGH 2	Formula	= least(100, greatest(0, 0.4 * (active_days_l30d / 30.0) * 100 + 0.3 * least(1, sessions_l30d / 60.0) * 100 + 0.3 * least(1, play_secs_l30d / 108000.0) * 100))	100.00

Column	Status and point	Kind	Rule	Pass (%)
frequency_decline	Affected HIGH 2	Formula	= CASE WHEN active_days_l30d > 0 THEN 100 * greatest(0, 1 - (active_days_l7d / 7.0) / (active_days_l30d / 30.0)) END	100.00
tp_avg_sessions_per_active_day	Affected HIGH 5	Limit	tp_avg_sessions_per_active_day >= 1 OR tp_avg_sessions_per_active_day = 0	99.47
tp_business_hours_ratio	Affected HIGH 5	Limit	tp_business_hours_ratio BETWEEN 0 AND 1	100.00
tp_max_sessions_single_day	Affected HIGH 5	Limit	tp_max_sessions_single_day >= 0	100.00
tp_window_days	Affected HIGH 5	No check	—	
risk_band	Affected MEDIUM 10	Band	bands of risk_score, cutoffs 25, 50, 75	
		Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
risk_score	Affected MEDIUM 10	Formula	= least(100, greatest(0, 20 * least(rg_loss_chasing_30d, 1) + 20 * least(rg_deposit_velocity, 1) + 15 * rg_night_play_ratio + 20 * least(1, greatest(0, stake_escalation)) + 25 * least(net_loss_velocity / 100, 1)))	100.00
session_intensity_score	Affected MEDIUM 8	Formula	= CASE WHEN active_days_l30d > 0 THEN least(100, 100 * play_secs_today / (play_secs_l30d / active_days_l30d + 1)) END	100.00
bet_z_score	Unconfirmed NO DEFINITION 25	Formula	$= (x - \bar{x}_{\text{brand}})/s_{\text{brand}}$ (M139)	0.00
ngr_eur_l180d	Unconfirmed NO DEFINITION 23	Identity	= ggr_eur_l180d - bonus_ggr_eur_l180d - tax_eur_l180d	78.90
ngr_eur_l30d	Unconfirmed NO DEFINITION 23	Window	= sum of ngr_eur_today over the 30 days up to D	99.73
		Identity	= ggr_eur_l30d - bonus_ggr_eur_l30d - tax_eur_l30d	92.70
ngr_eur_l7d	Unconfirmed NO DEFINITION 23	Window	= sum of ngr_eur_today over the 7 days up to D	99.98
		Identity	= ggr_eur_l7d - bonus_ggr_eur_l7d - tax_eur_l7d	96.35

Column	Status and point	Kind	Rule	Pass (%)
ngr_eur_l90d	Unconfirmed NO DEFINITION 23	Identity	= ggr_eur_l90d - bonus_ggr_eur_l90d - tax_eur_l90d	86.94
ngr_eur_today	Unconfirmed NO DEFINITION 23	Identity	= ggr_eur_today - bonus_ggr_eur_today - tax_eur_today	98.84
prior_betting_aggression_score	Unconfirmed NO DEFINITION 24	Lag	= betting_aggression_score, 7 days before	49.36
prior_churn_score	Unconfirmed NO DEFINITION 24	Lag	= churn_score, 7 days before	84.04
prior_engagement_score	Unconfirmed NO DEFINITION 24	Lag	= engagement_score, 7 days before	85.26
prior_risk_score	Unconfirmed NO DEFINITION 24	Lag	= risk_score, 7 days before	92.98
session_len_z	Unconfirmed NO DEFINITION 25	Formula	$= (x - \bar{x}_{\text{brand}})/s_{\text{brand}}$ (M136)	0.32
avg_bet_eur_l30d	Pass	Identity	= CASE WHEN bets_l30d > 0 THEN wagered_eur_l30d / bets_l30d END	100.00
avg_bet_eur_l7d	Pass	Identity	= CASE WHEN bets_l7d > 0 THEN wagered_eur_l7d / bets_l7d END	100.00
bets_l180d	Pass	Limit	bets_l180d >= bets_l90d	100.00
bets_l30d	Pass	Window Limit	= sum of bets_today over the 30 days up to D bets_l30d >= bets_l7d	99.97 100.00
bets_l7d	Pass	Window Limit	= sum of bets_today over the 7 days up to D bets_l7d >= bets_today	100.00 100.00
bets_l90d	Pass	Limit	bets_l90d >= bets_l30d	100.00
bets_today	Pass	Limit	bets_today >= 0	100.00
betting_aggression_score	Pass	Formula	= least(100, greatest(0, 50 + 25 * least(2, greatest(-2, bet_z_score)) + 25 * least(1, greatest(-1, stake_escalation))))	100.00
bonus_ggr_eur_l180d	Pass	Identity	= ggr_eur_l180d - real_ggr_eur_l180d	100.00
bonus_ggr_eur_l30d	Pass	Window	= sum of bonus_ggr_eur_today over the 30 days up to D	99.98
bonus_ggr_eur_l7d	Pass	Window	= sum of bonus_ggr_eur_today over the 7 days up to D	99.99
bonus_ggr_eur_l90d	Pass	Identity	= ggr_eur_l90d - real_ggr_eur_l90d	100.00
bonus_ggr_eur_today	Pass	Identity	= ggr_eur_today - real_ggr_eur_today	100.00

Column	Status and point	Kind	Rule	Pass (%)
		Limit	bonus_ggr_eur_today <= wagered_bonus_eur_today + 0.01	100.00
		Validated	= sum of bonus_ggr_eur of gld_player_gaming_daily for the player and day (% of the player-days with play)	99.92
bonus_granted_eur_l30d	Pass	Limit	bonus_granted_eur_l30d >= 0	100.00
bonus_wagered_eur_l30d	Pass	Limit	bonus_wagered_eur_l30d >= 0	100.00
cashout_eur_l180d	Pass	Limit	cashout_eur_l180d >= cashout_eur_l90d	100.00
cashout_eur_l30d	Pass	Window	= sum of cashout_eur_today over the 30 days up to D	100.00
		Limit	cashout_eur_l30d >= cashout_eur_l7d	100.00
cashout_eur_l7d	Pass	Window	= sum of cashout_eur_today over the 7 days up to D	100.00
		Limit	cashout_eur_l7d >= cashout_eur_today	100.00
cashout_eur_l90d	Pass	Limit	cashout_eur_l90d >= cashout_eur_l30d	100.00
cashout_eur_today	Pass	Limit	cashout_eur_today >= 0	100.00
churn_band_changed_at	Pass	Limit	churn_band_changed_at <= snapshot_date	100.00
confidence_score	Pass	Limit	confidence_score BETWEEN 0 AND 1	100.00
current_active_streak_days	Pass	Limit	current_active_streak_days <= tenure_days + 1	100.00
deposits_eur_l180d	Pass	Limit	deposits_eur_l180d >= deposits_eur_l90d	100.00
deposits_eur_l30d	Pass	Window	= sum of deposits_eur_today over the 30 days up to D	99.99
		Limit	deposits_eur_l30d >= deposits_eur_l7d	100.00
deposits_eur_l7d	Pass	Window	= sum of deposits_eur_today over the 7 days up to D	99.99
		Limit	deposits_eur_l7d >= deposits_eur_today	100.00
deposits_eur_l90d	Pass	Limit	deposits_eur_l90d >= deposits_eur_l30d	100.00
deposits_eur_today	Pass	Limit	deposits_eur_today >= 0	100.00
games_breadth_l30d	Pass	Limit	games_breadth_l30d >= 0	100.00
gap_days	Pass	Limit	gap_days >= 0	100.00
ggr_eur_l180d	Pass	Identity	= wagered_eur_l180d - won_eur_l180d	100.00
ggr_eur_l30d	Pass	Window	= sum of ggr_eur_today over the 30 days up to D	99.93

Column	Status and point	Kind	Rule	Pass (%)
		Identity	= `wagered_eur_l30d - won_eur_l30d`	100.00
`ggr_eur_l7d`	Pass	Window	= sum of `ggr_eur_today` over the 7 days up to D	99.98
		Identity	= `wagered_eur_l7d - won_eur_l7d`	100.00
`ggr_eur_l90d`	Pass	Identity	= `wagered_eur_l90d - won_eur_l90d`	100.00
`ggr_eur_today`	Pass	Identity	= `wagered_eur_today - won_eur_today`	100.00
`lifecycle_stage`	Pass	Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
`lifecycle_stage_changed_at`	Pass	Limit	`lifecycle_stage_changed_at <= snapshot_date`	100.00
`lifetime_value_tier`	Pass	Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
`net_cash_eur_l180d`	Pass	Identity	= `deposits_eur_l180d - withdrawals_eur_l180d`	100.00
`net_cash_eur_l30d`	Pass	Window	= sum of `net_cash_eur_today` over the 30 days up to D	99.97
		Identity	= `deposits_eur_l30d - withdrawals_eur_l30d`	100.00
`net_cash_eur_l7d`	Pass	Window	= sum of `net_cash_eur_today` over the 7 days up to D	99.99
		Identity	= `deposits_eur_l7d - withdrawals_eur_l7d`	100.00
`net_cash_eur_l90d`	Pass	Identity	= `deposits_eur_l90d - withdrawals_eur_l90d`	100.00
`net_cash_eur_today`	Pass	Identity	= `deposits_eur_today - withdrawals_eur_today`	100.00
`net_loss_velocity`	Pass	Formula	= `CASE WHEN deposits_eur_l30d > 0 THEN greatest(0, -ggr_eur_today) / (deposits_eur_l30d / 30 + 1) ELSE 0 END`	100.00
`player_id`	Pass	Format	`regexp_full_match( CAST(player_id AS VARCHAR), '[0-9]{5, 8}')`	100.00
`prev_churn_band`	Pass	Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
`prev_lifecycle_stage`	Pass	Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
`prev_risk_band`	Pass	Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
`prev_value_tier`	Pass	Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42

Column	Status and point	Kind	Rule	Pass (%)
prior_avg_bet_eur_l7d	Pass	Identity	= CASE WHEN prior_bets_l7d > 0 THEN prior_wagered_eur_l7d / prior_bets_l7d END	100.00
prior_bets_l30d	Pass	Lag	= bets_l30d, 30 days before	100.00
prior_bets_l7d	Pass	Lag	= bets_l7d, 7 days before	100.00
prior_deposits_eur_l30d	Pass	Lag	= deposits_eur_l30d, 30 days before	100.00
prior_deposits_eur_l7d	Pass	Lag	= deposits_eur_l7d, 7 days before	100.00
prior_ggr_eur_l30d	Pass	Lag	= ggr_eur_l30d, 30 days before	100.00
prior_ggr_eur_l7d	Pass	Lag	= ggr_eur_l7d, 7 days before	100.00
prior_wagered_eur_l30d	Pass	Lag	= wagered_eur_l30d, 30 days before	100.00
prior_wagered_eur_l7d	Pass	Lag	= wagered_eur_l7d, 7 days before	100.00
real_ggr_eur_l180d	Pass	Identity	= ggr_eur_l180d - bonus_ggr_eur_l180d	100.00
real_ggr_eur_l30d	Pass	Window	= sum of real_ggr_eur_today over the 30 days up to D	99.93
		Identity	= ggr_eur_l30d - bonus_ggr_eur_l30d	100.00
real_ggr_eur_l7d	Pass	Window	= sum of real_ggr_eur_today over the 7 days up to D	99.98
		Identity	= ggr_eur_l7d - bonus_ggr_eur_l7d	100.00
real_ggr_eur_l90d	Pass	Identity	= ggr_eur_l90d - bonus_ggr_eur_l90d	100.00
real_ggr_eur_today	Pass	Identity	= ggr_eur_today - bonus_ggr_eur_today	100.00
rg_deposit_velocity	Pass	Formula	= CASE WHEN deposits_eur_l30d > 0 THEN deposits_eur_l7d / (deposits_eur_l30d / 30 * 7) ELSE 0 END	100.00
rg_loss_chasing_30d	Pass	Formula	= CASE WHEN wagered_eur_l30d > 0 AND ggr_eur_l30d < 0 THEN least(1, -ggr_eur_l30d / wagered_eur_l30d) ELSE 0 END	100.00
risk_appetite_score	Pass	Formula	= least(100, greatest(0, 50 + 50 * least(1, greatest(-1, bet_z_score))))	100.00
risk_band_changed_at	Pass	Limit	risk_band_changed_at <= snapshot_date	100.00
sessions_l180d	Pass	Limit	sessions_l180d >= sessions_l90d	100.00
sessions_l30d	Pass	Window	= sum of sessions_today over the 30 days up to D	99.98

Column	Status and point	Kind	Rule	Pass (%)
		Limit	sessions_l30d >= sessions_l7d	100.00
sessions_l7d	Pass	Window	= sum of sessions_today over the 7 days up to D	100.00
		Limit	sessions_l7d >= sessions_today	100.00
sessions_l90d	Pass	Limit	sessions_l90d >= sessions_l30d	100.00
sessions_today	Pass	Limit	sessions_today >= 0	100.00
sessions_with_bonus_l30d	Pass	Limit	sessions_with_bonus_l30d <= sessions_l30d	100.00
stake_escalation	Pass	Formula	= CASE WHEN avg_bet_eur_l30d > 0 THEN (avg_bet_eur_l7d - avg_bet_eur_l30d) / avg_bet_eur_l30d ELSE 0 END	100.00
tenant_id	Pass	Format	regexp_full_match(tenant_id, '[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}')	100.00
theo_win_eur_l30d	Pass	Limit	theo_win_eur_l30d >= 0 AND theo_win_eur_l30d <= wagered_eur_l30d	100.00
value_rank_within_tenant	Pass	Limit	value_rank_within_tenant >= 1	100.00
value_score_eur	Pass	Formula	= greatest(0, ggr_eur_l7d + 0.70 * (ggr_eur_l30d - ggr_eur_l7d) + 0.40 * (ggr_eur_l90d - ggr_eur_l30d) + 0.20 * (ggr_eur_l180d - ggr_eur_l90d))	100.00
value_tier	Pass	Band	bands of value_score_eur, no cutoffs in the workbook	
		Distribution	not uniform; if it has levels, they fall from the first to the last	Fig. 42
value_tier_changed_at	Pass	Limit	value_tier_changed_at <= snapshot_date	100.00
wagered_bonus_eur_l30d	Pass	Limit	wagered_bonus_eur_l30d >= 0	100.00
		Window	= sum of wagered_bonus_eur_today over the 30 days up to D	99.97
wagered_bonus_eur_today	Pass	Limit	wagered_bonus_eur_today >= 0	100.00
wagered_eur_l180d	Pass	Limit	wagered_eur_l180d >= wagered_eur_l90d	100.00
wagered_eur_l30d	Pass	Window	= sum of wagered_eur_today over the 30 days up to D	99.96

Column	Status and point	Kind	Rule	Pass (%)
		Limit	`wagered_eur_l30d >= wagered_eur_l7d`	100.00
		Identity	= `wagered_real_eur_l30d + wagered_bonus_eur_l30d`	100.00
`wagered_eur_l7d`	Pass	Window	= sum of `wagered_eur_today` over the 7 days up to D	99.99
		Limit	`wagered_eur_l7d >= wagered_eur_today`	100.00
`wagered_eur_l90d`	Pass	Limit	`wagered_eur_l90d >= wagered_eur_l30d`	100.00
`wagered_eur_today`	Pass	Limit	`wagered_eur_today >= 0`	100.00
		Identity	= `wagered_real_eur_today + wagered_bonus_eur_today`	100.00
`wagered_real_eur_l30d`	Pass	Limit	`wagered_real_eur_l30d >= 0`	100.00
		Window	= sum of `wagered_real_eur_today` over the 30 days up to D	99.96
`wagered_real_eur_today`	Pass	Limit	`wagered_real_eur_today >= 0`	100.00
`withdrawals_eur_l180d`	Pass	Limit	`withdrawals_eur_l180d >= withdrawals_eur_l90d`	100.00
`withdrawals_eur_l30d`	Pass	Window	= sum of `withdrawals_eur_today` over the 30 days up to D	99.99
		Limit	`withdrawals_eur_l30d >= withdrawals_eur_l7d`	100.00
`withdrawals_eur_l7d`	Pass	Window	= sum of `withdrawals_eur_today` over the 7 days up to D	100.00
		Limit	`withdrawals_eur_l7d >= withdrawals_eur_today`	100.00
`withdrawals_eur_l90d`	Pass	Limit	`withdrawals_eur_l90d >= withdrawals_eur_l30d`	100.00
`withdrawals_eur_today`	Pass	Limit	`withdrawals_eur_today >= 0`	100.00
`won_eur_l180d`	Pass	Limit	`won_eur_l180d >= won_eur_l90d`	100.00
`won_eur_l30d`	Pass	Window	= sum of `won_eur_today` over the 30 days up to D	99.97
		Limit	`won_eur_l30d >= won_eur_l7d`	100.00
`won_eur_l7d`	Pass	Window	= sum of `won_eur_today` over the 7 days up to D	99.99
		Limit	`won_eur_l7d >= won_eur_today`	100.00
`won_eur_l90d`	Pass	Limit	`won_eur_l90d >= won_eur_l30d`	100.00
`won_eur_today`	Pass	Limit	`won_eur_today >= 0`	100.00
`_ingested_at`	Not checkable	No check	—	

Column	Status and point	Kind	Rule	Pass (%)
`_processed_at`	**Not checkable**	No check	—	
`_version`	**Not checkable**	No check	—	
`has_sufficient_data`	**Not checkable**	No check	—	
`param_version`	**Not checkable**	No check	—	
`snapshot_date`	**Not checkable**	No check	—	